maelcum mixer — readable schematics of the generated boards

One page per board. Each drawing is generated from the same netlist as the board's PCB
and checked against it in both directions (pcb/tools/leggibile.py). Values are PROVISIONAL drafts.

Page	Board
2	maelcum CHANNEL CDJ/LINE (v0.4) — draft v0.1 (383 parts, rev. 0.2-draft)
3	maelcum CHANNEL control board — draft v0.1 (34 parts, rev. 0.1-draft)
4	maelcum FX-SEND (dec. 91a) — draft v0.2 (120 parts, rev. 0.3-draft)
5	maelcum FX-SEND control board — draft v0.2 (4 parts, rev. 0.2-draft)
6	maelcum FX-RET (dec. 130) — draft v0.1 (199 parts, rev. 0.1-draft)
7	maelcum FX-RET reroute (dec. 91) — draft v0.1 (225 parts, rev. 0.1-draft)
8	maelcum FX-RET control board — draft v0.1 (23 parts, rev. 0.1-draft)
9	maelcum MASTER signal board (dec. 140) — draft v0.1 (285 parts, rev. 0.2-draft)
10	maelcum MASTER OUTPUT board L (dec. 142-143) — draft v0.1 (155 parts, rev. 0.1-draft)
11	maelcum MASTER OUTPUT board R (dec. 142-143) — draft v0.1 (144 parts, rev. 0.1-draft)
12	maelcum MASTER control board — draft v0.1 (60 parts, rev. 0.1-draft)
13	maelcum MIC (dec. 97) — draft v0.1 (213 parts, rev. 0.1-draft)
14	maelcum MIC control board — draft v0.1 (10 parts, rev. 0.1-draft)
15	maelcum MONO (dec. 97) — draft v0.1 (127 parts, rev. 0.1-draft)
16	maelcum MONO control board — draft v0.1 (9 parts, rev. 0.1-draft)
17	maelcum CUE / cuffie (sistema-cue v0.2) — draft v0.1 (160 parts, rev. 0.1-draft)
18	maelcum CUE control board — draft v0.1 (6 parts, rev. 0.1-draft)
19	maelcum PSU-IN (dec. 113) — draft v0.1 (41 parts, rev. 0.1-draft)
20	maelcum backplane T1 (rear: MASTER, 5-WAY, PSU-IN) — draft v0.1 (75 parts, rev. 0.1-draft)
21	maelcum backplane T2 (channels: FILTER, 4 CHANNELS, 4 FX-SEND) — draft v0.1 (170 parts, rev. 0.1-draft)
22	maelcum backplane T3 (headphones: CUE) — draft v0.1 (22 parts, rev. 0.1-draft)
23	maelcum backplane T4 (right: MIC, 3 MONO, 4 FX-RET) — draft v0.1 (137 parts, rev. 0.1-draft)

How to read the drawings

- Arrow and ground symbols are power rails and grounds; the name under the symbol is the rail (AGND, +18V, -15SW ...).
- Resistors and capacitors are wired to the op-amp pin they touch: in series along the pin, shunt towards the rail.
- A plain name on a wire is a net that continues elsewhere in the same block.
- A name in a box is a net that goes to another block on the same board, or to a connector.
- Connectors and many-pin chips (C96, LM339, ADG1434) carry only names; their parts are drawn in chains below them.
- Dashed frames are the functional blocks of the board, in signal order. Net names are the PCB identifiers.

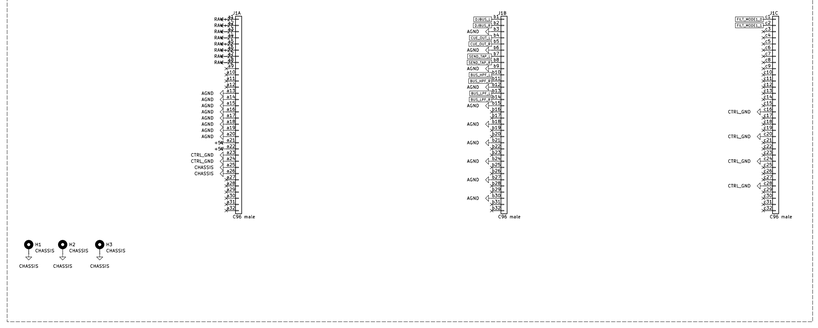
Generated by pcb/tools/leggibile.py

Sheet: /
File: schematics-cover.kicad_sch

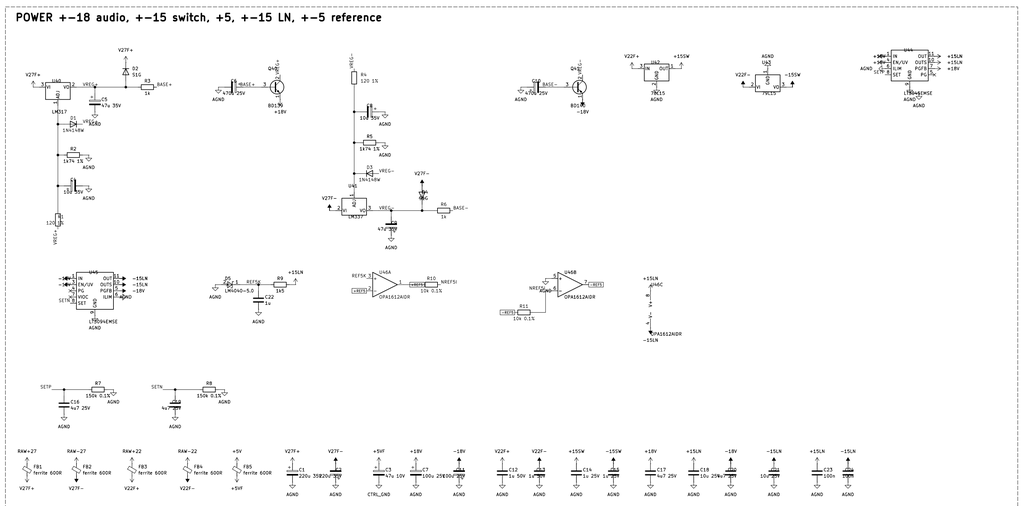
Title: maelcum mixer — readable schematics

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 1/1

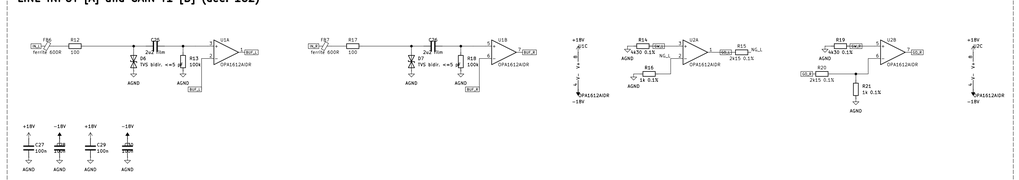
BACKPLANE C96 (CDJ class, rev. dec. 159)



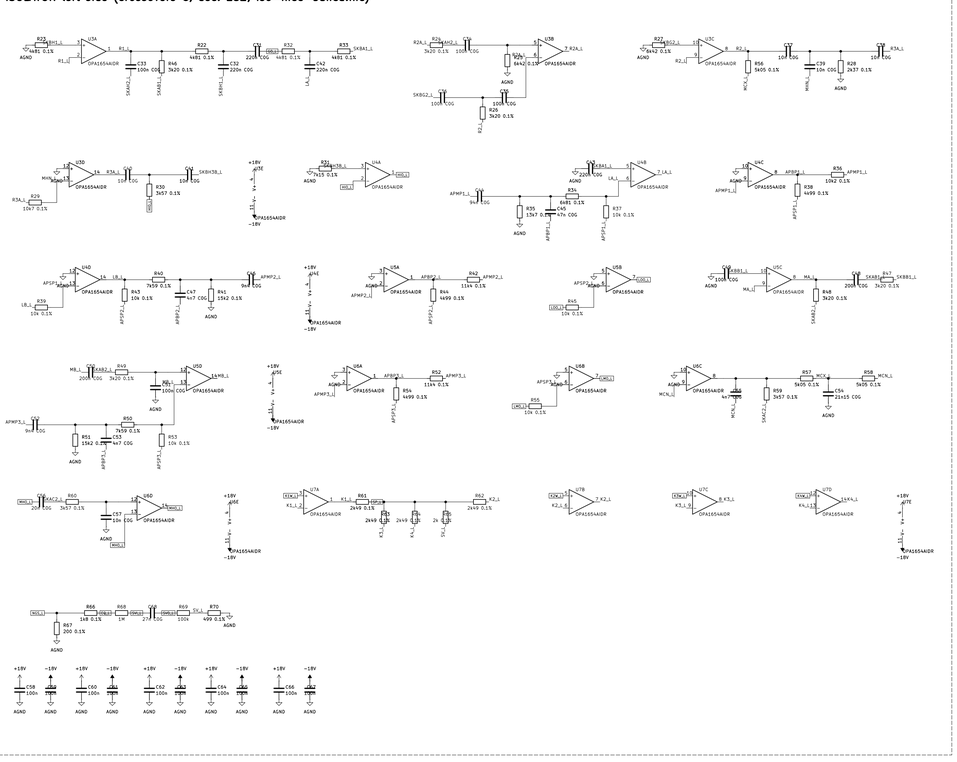
POWER +−18 audio, +−15 switch, +5, +−15 LN, +−5 reference



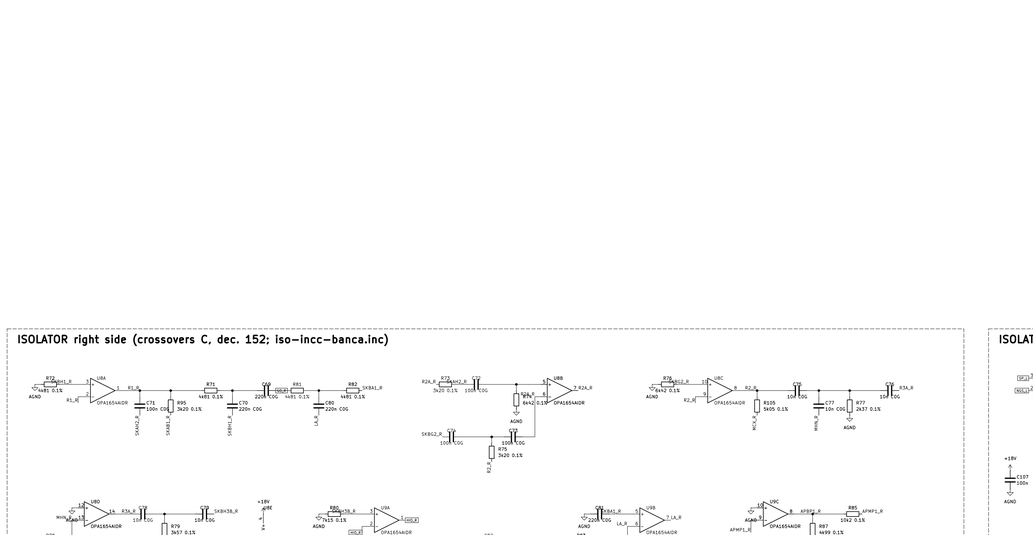
LINE INPUT [A] and GAIN T1 [B] (dec. 162)



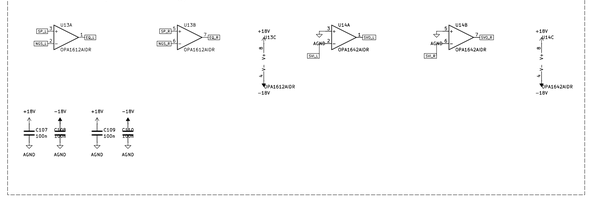
ISOLATOR left side (crossovers C, dec. 152; iso−incc−banca.inc)



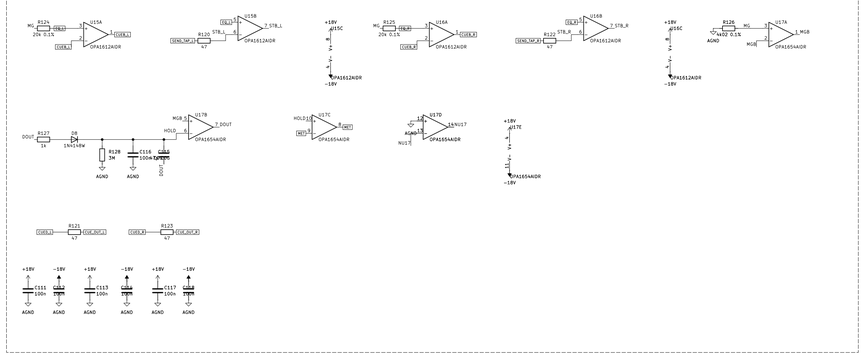
FADER [H] (2 VCAs, fader shaper, dec. 126) and DIP (dec. 163)



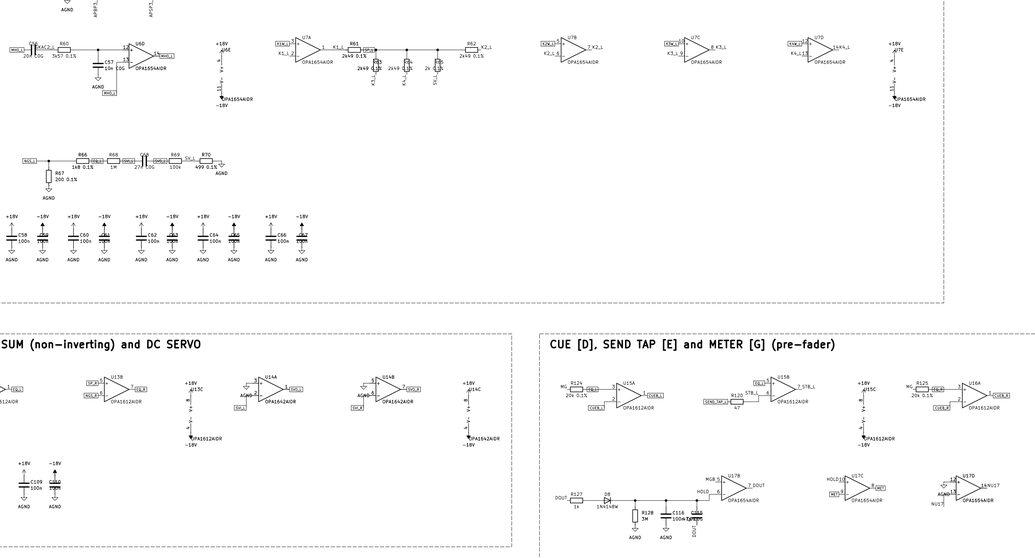
ISOLATOR SUM (non−inverting) and DC SERVO



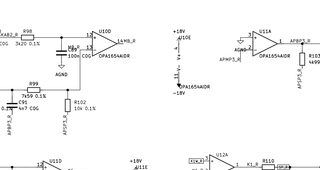
CUE [D], SEND TAP [E] and METER [G] (pre−fader)



HPF/OFF/LPF TOGGLE: self−timed logic (74HC14, 74HC86, 74HC74) and ADG1434



CONTROL BOARD CABLE (J2) and RCA INPUTS (J3)



PWR_FLAG (for the ERC check only)



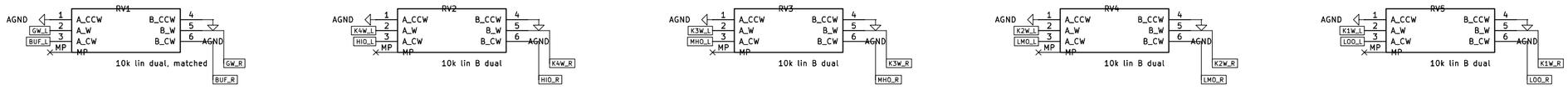
Generated by pcb/tools/leggible.py from the same netlist as the PCB; do not edit by hand

Sheet: /
File: iparate_bloqd_sch

Title: maelcum CHANNEL CDJ/LINE (v0.4) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.2−draft
KiCad E.D.A. 10.0.7		Id: 1/1

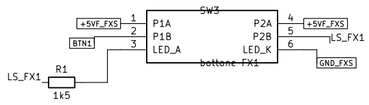
POTS: GAIN and ISOLATOR (dual pots, section A = L, B = R)

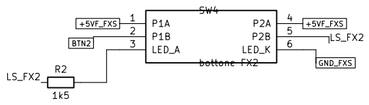


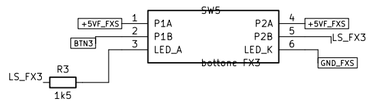
TOGGLES: FILTER (HPF/OFF/LPF) and LINE/PHONO

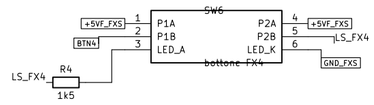


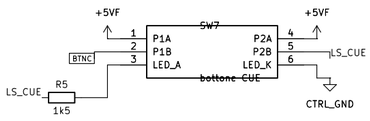
BUTTONS: FX1–4 (assign to the FX–SEND) and CUE



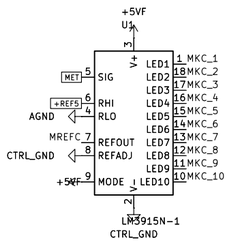


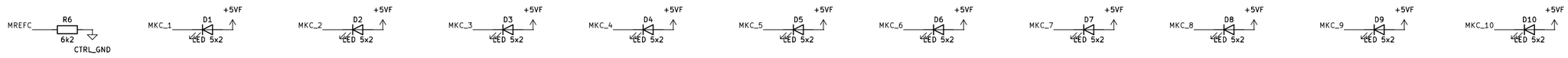


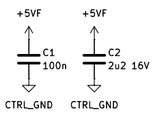




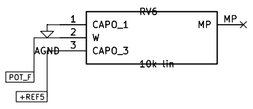
METER: one LM3915, 10 LEDs (+24…–3 dBu)



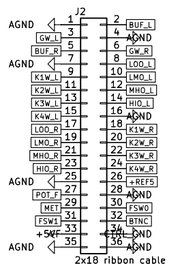


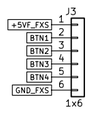


FADER (PTA6043, 60 mm travel, dec. 169)

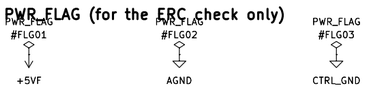


CABLES: J2 to canale.py (J2), J3 to fx–send.py (J3)





PWR_FLAG (for the ERC check only)



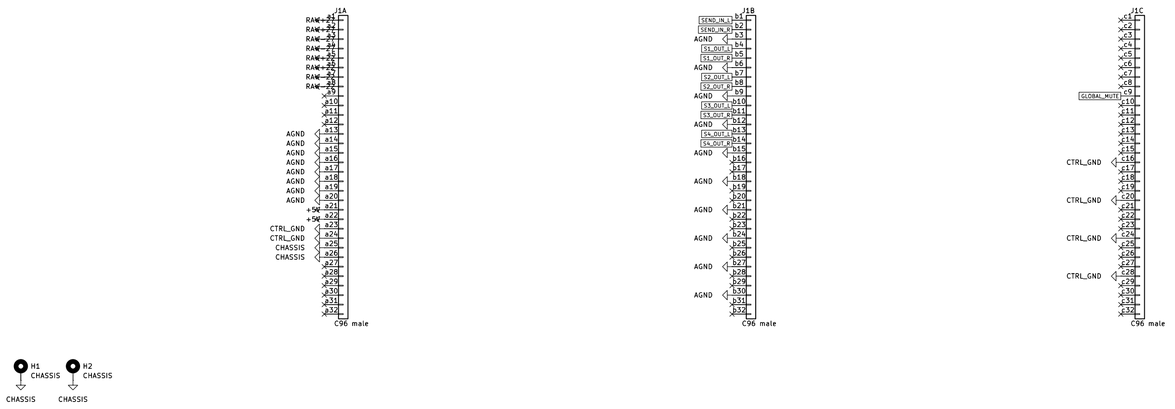
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: canale–comandi.kicad_sch

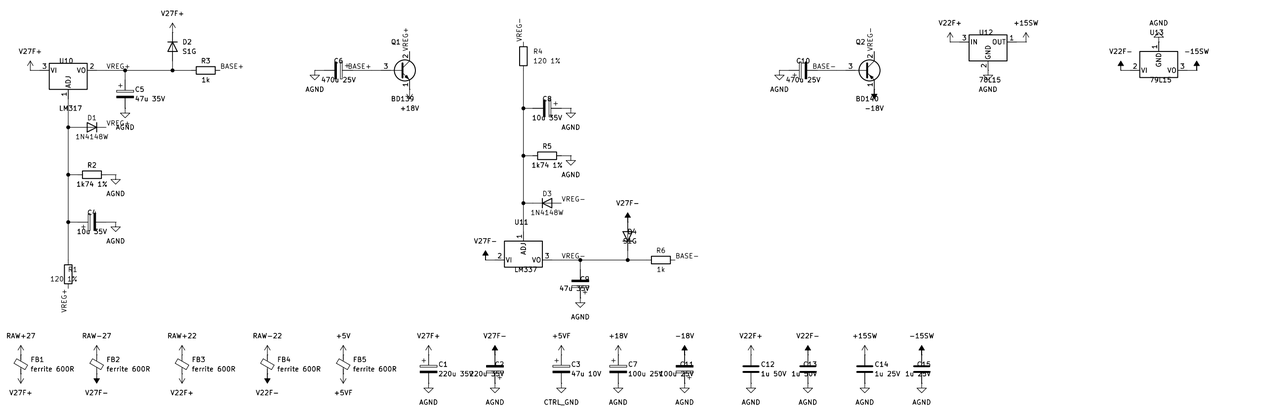
Title: maelcum CHANNEL control board – draft v0.1 – readable schematic

Size: User	Date:	Rev: 0.1–draft
KiCad E.D.A. 10.0.7		Id: 1/1

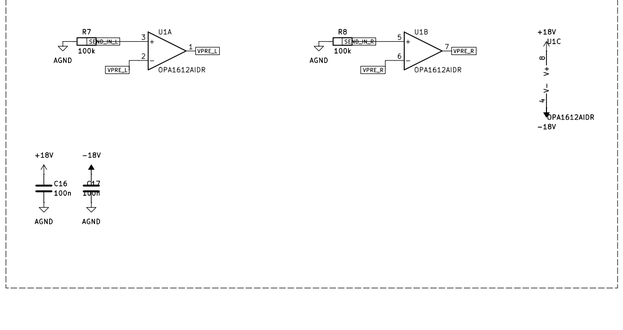
BACKPLANE C96 (FX-SEND class)



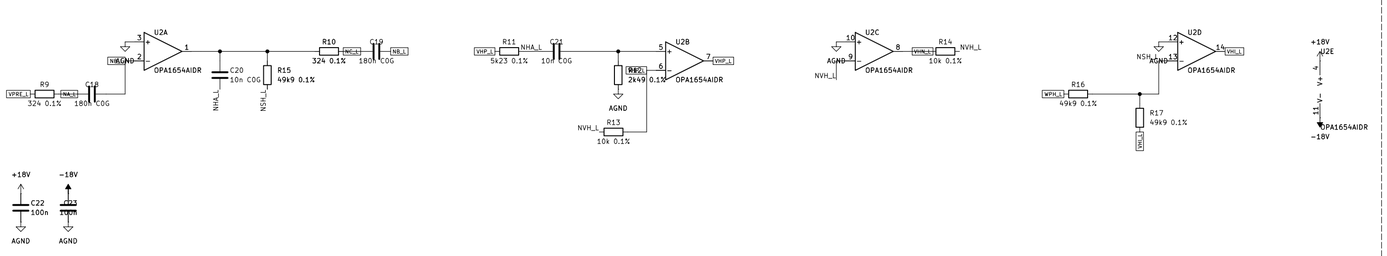
POWER +-18 audio, +-15 switch, +5



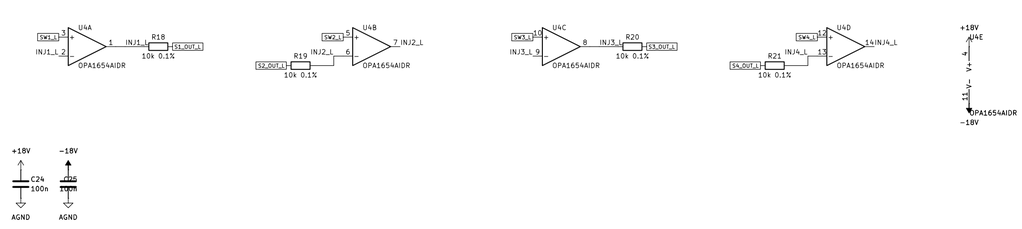
INPUT BUFFER (OPA1612)



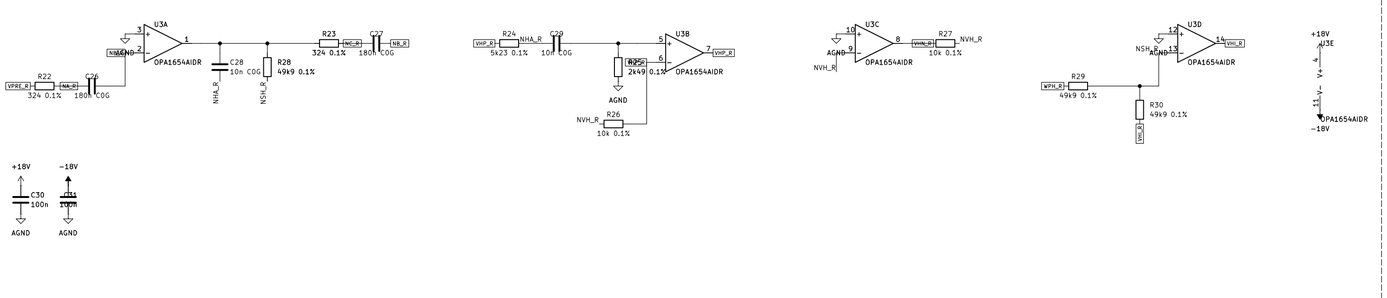
EQ2 left side (LOW 120 Hz + HIGH 4.5 kHz, fx-eq2-pot10k.cir)



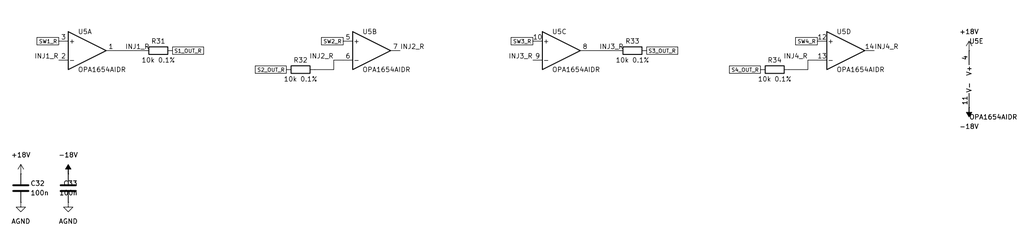
INJECTION left side (OPA1654 + 10k to the buses)



EQ2 right side (LOW 120 Hz + HIGH 4.5 kHz, fx-eq2-pot10k.cir)



INJECTION right side (OPA1654 + 10k to the buses)



ASSIGN S1-S4 (ADG1434 pi SPDT, 1 ms soft switch, global mute)

CONTROL BOARD CABLE (J2) and ASSIGN BUTTONS (J3)

PWR_FLAG (for the ERC check only)

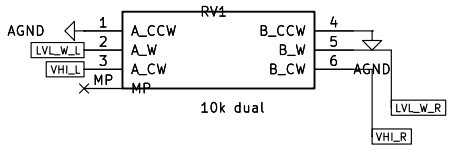
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

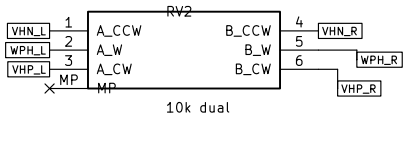
Sheet: /
File: fx-send.kicad_sch

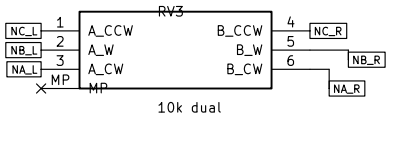
Title: maelcum FX-SEND (dec. 91a) — draft v0.2 — readable schematic

Size: User | Date: | Rev: 0.3-draft
KiCad E.D.A. 10.0.7 | Id: 1/1

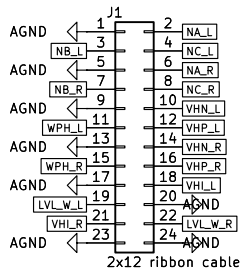
POTS (positions from the panel render)



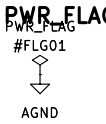




CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)



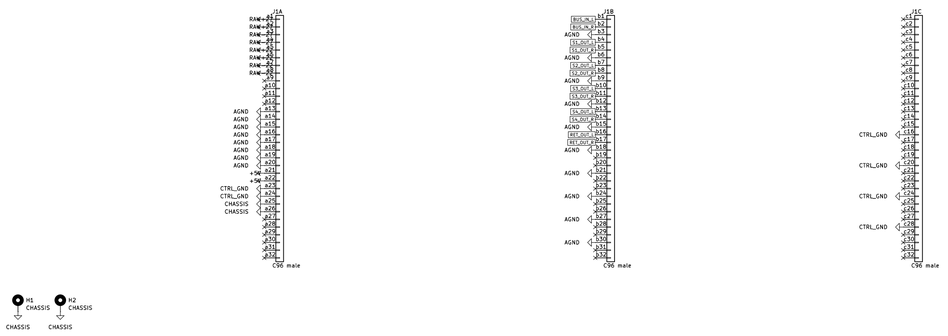
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: fx-send-comandi.kicad_sch

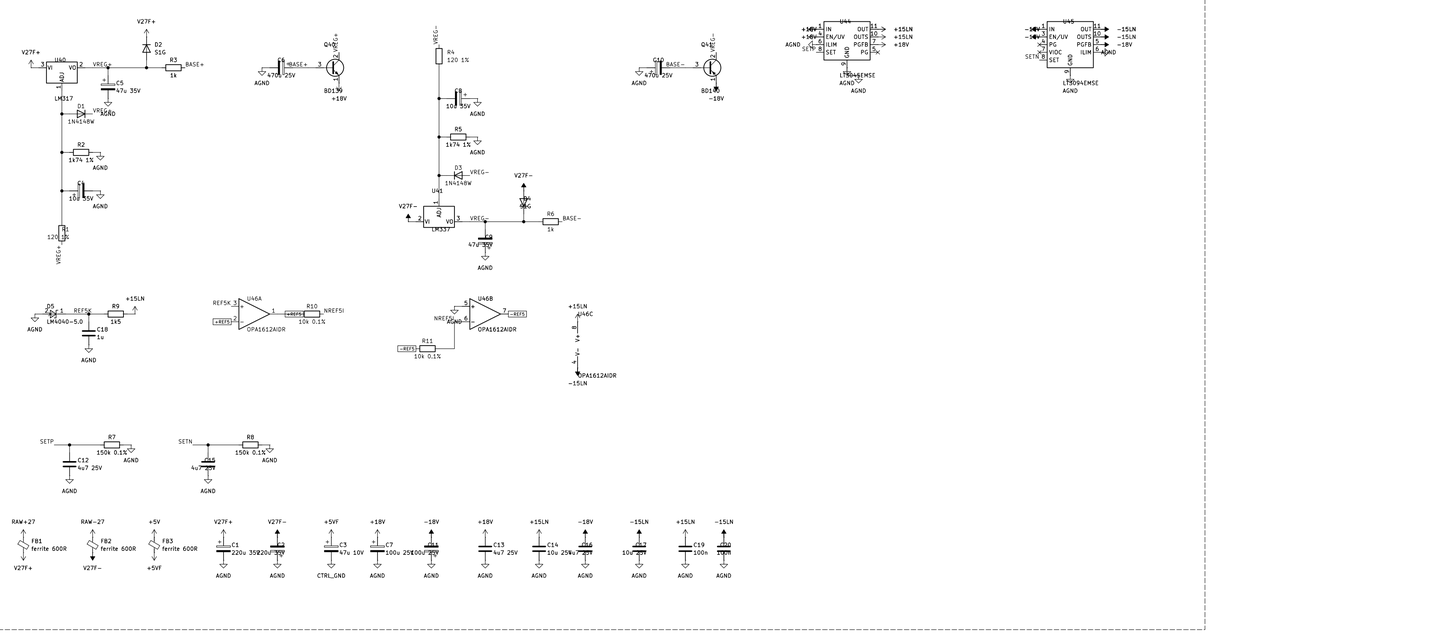
Title: maelcum FX-SEND control board — draft v0.2 — readable schematic

Size: User	Date:	**Rev: 0.2-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

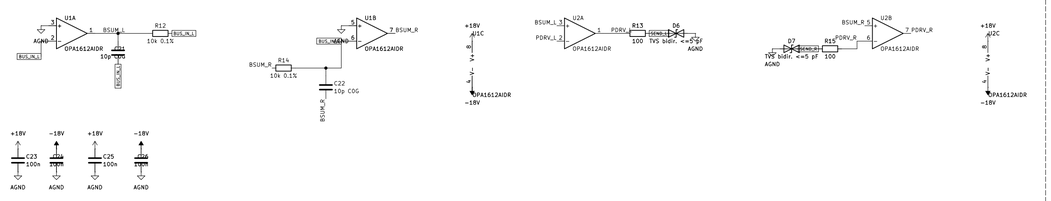
BACKPLANE C96 (FX-RET class, dec. 130)



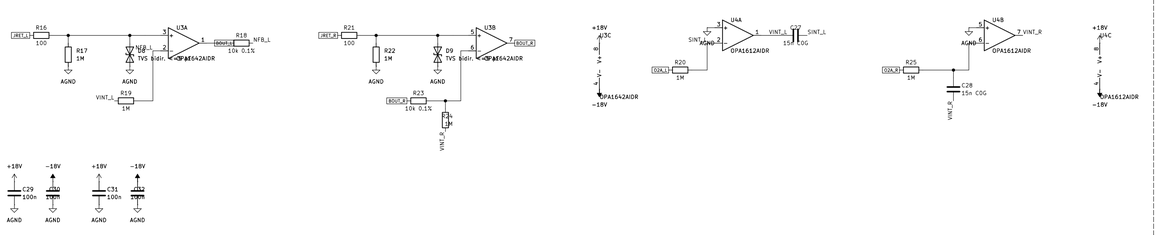
POWER +-18 audio, +5, +-15 LN, +-5 reference



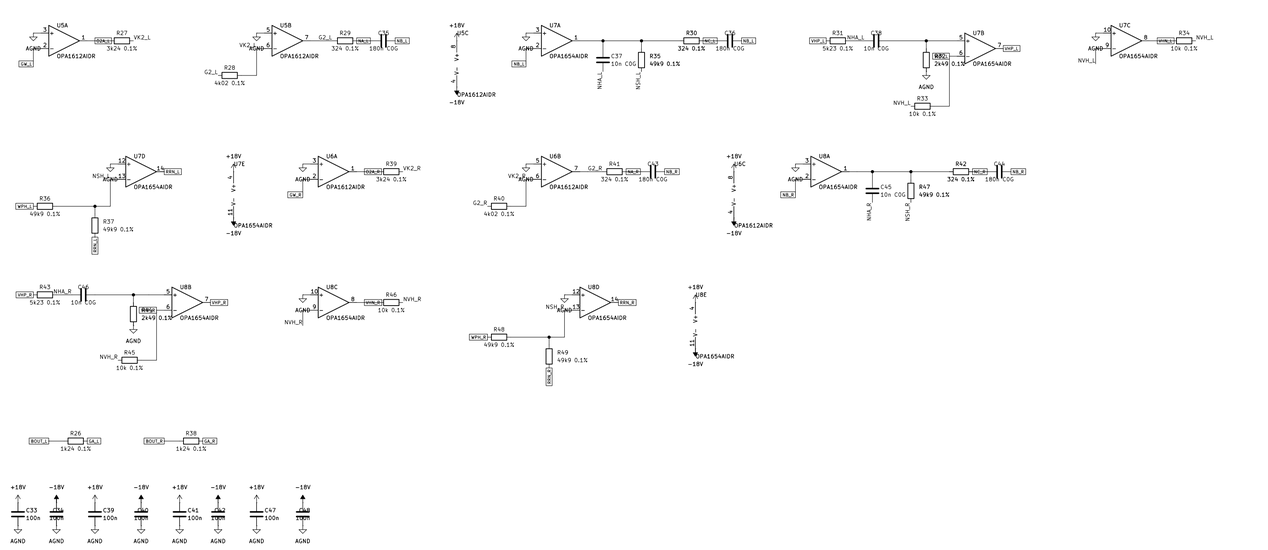
BUS SUM (slot 1, Rf 10k, Cf 10p) and PEDAL DRIVER



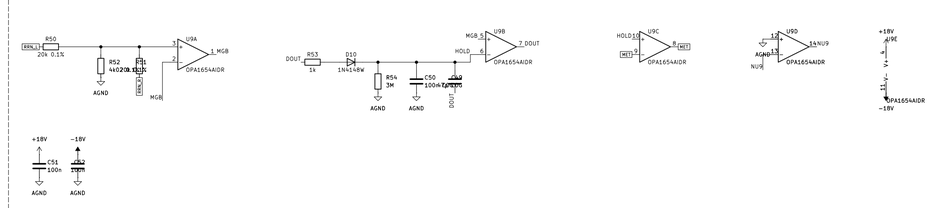
RETURN: protection, JFET buffer g=1, DC servo (ritorno-buffer-g1.cir)



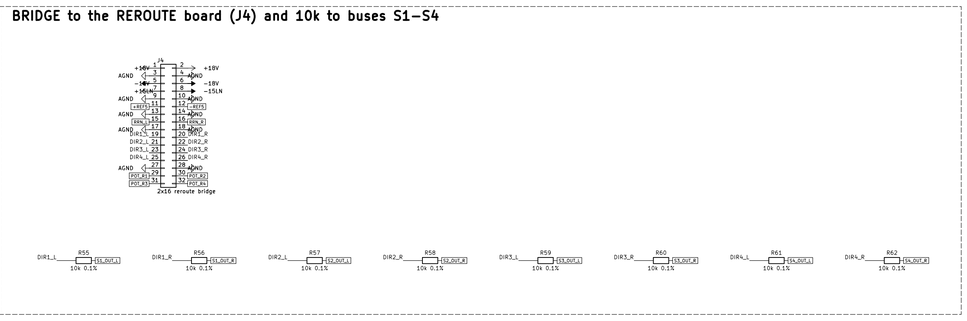
GAIN T2 (dual 10k lin pot on the cable, dec. 162; x1.24 todo 220) and EQ2



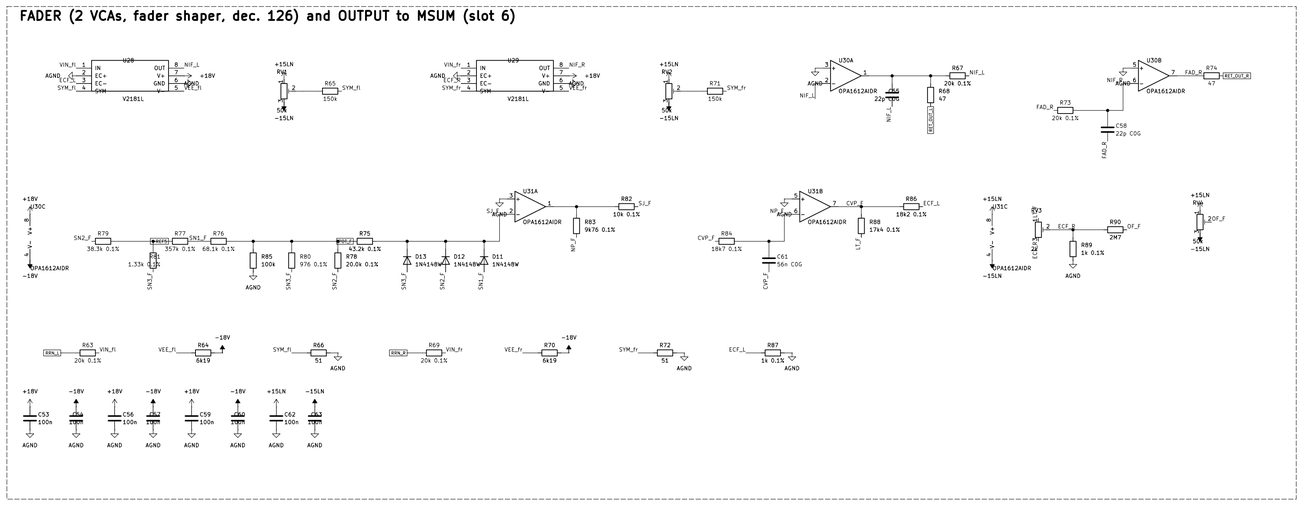
RETURN METER (centred mono, g = 0.288; detector meter-det-03s.cir)



BRIDGE to the REROUTE board (J4) and 10k to buses S1-S4



FADER (2 VCAs, fader shaper, dec. 126) and OUTPUT to MSUM (slot 6)



CONTROL BOARD CABLE (J2) and PEDAL JACK (J3)



PWR_FLAG (for the ERC check only)



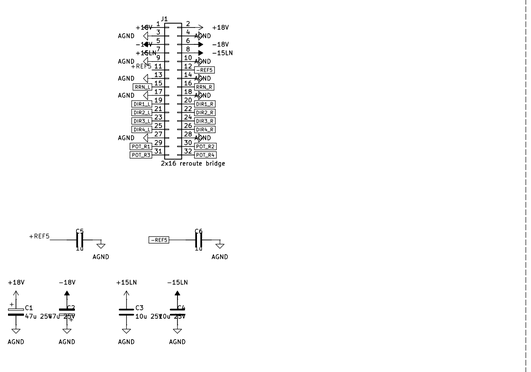
Generated by pcb/tools/leggible.py from the same netlist as the PCB: do not edit by hand.

Sheet: /
File: fx-ret.kicad_sch

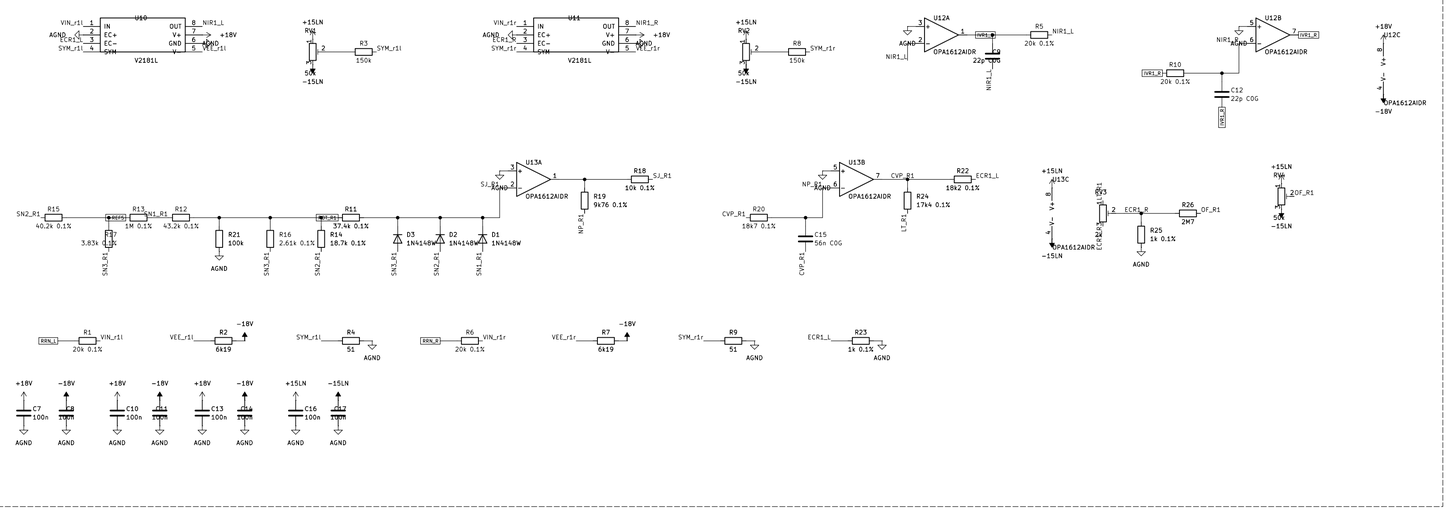
Title: maelcum FX-RET (dec. 130) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

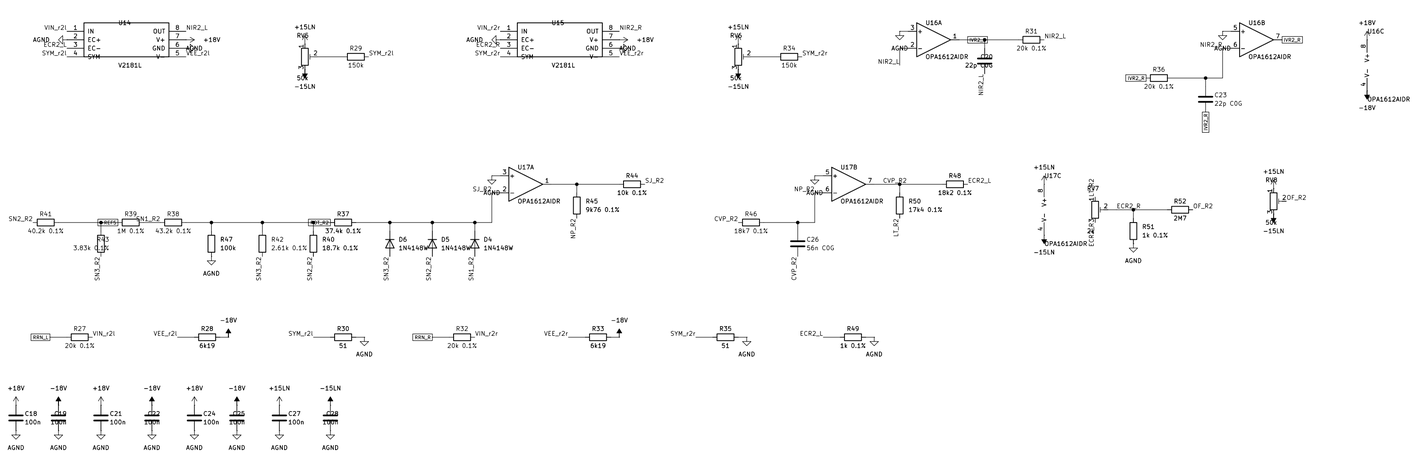
BRIDGE from the signal board (J1) and local bulk caps



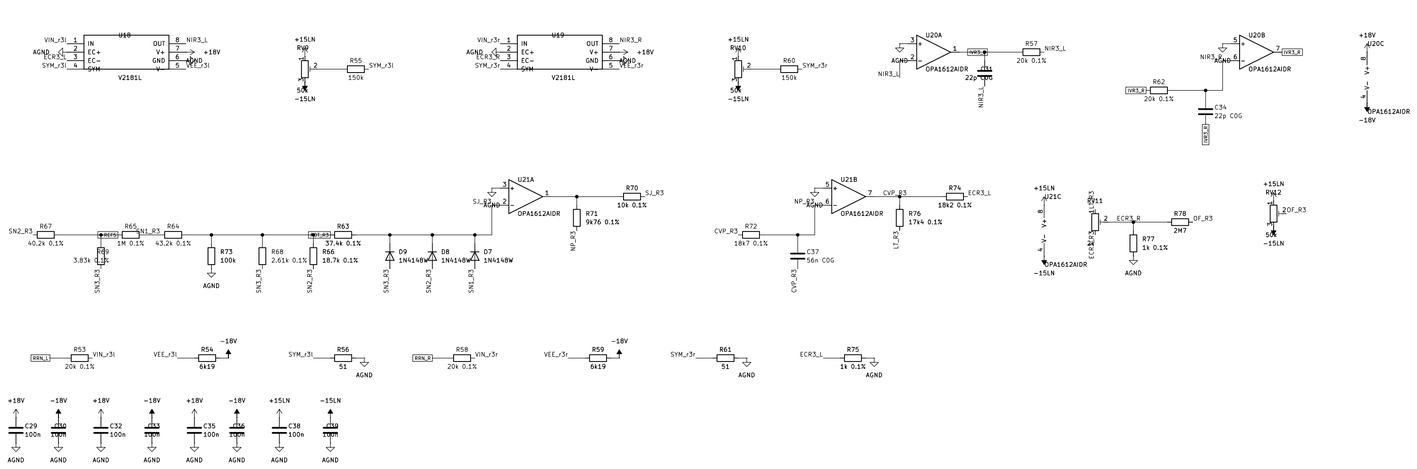
REROUTE to S1 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



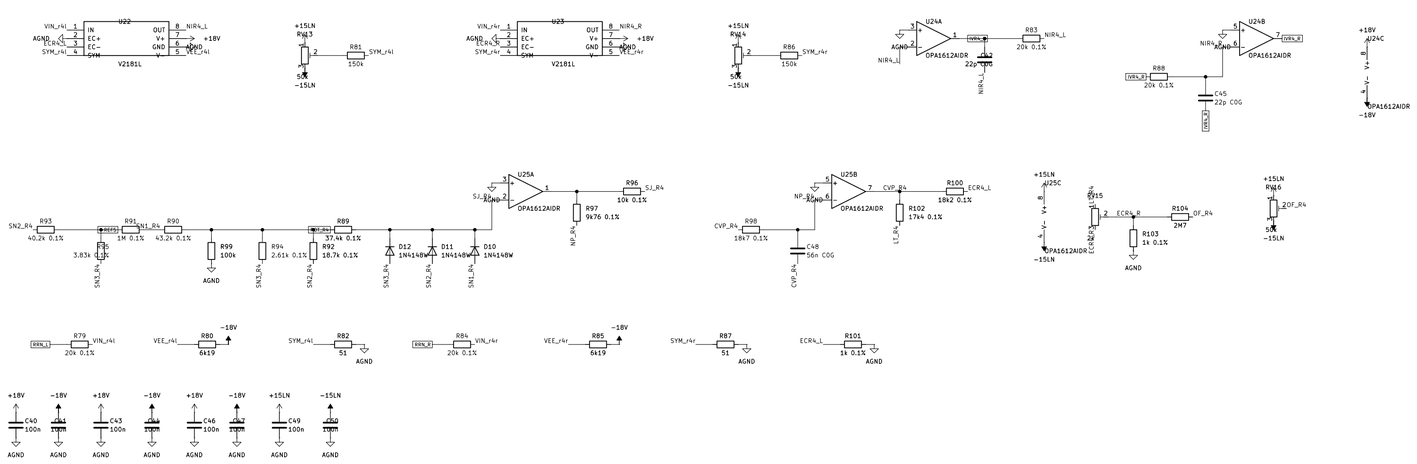
REROUTE to S2 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



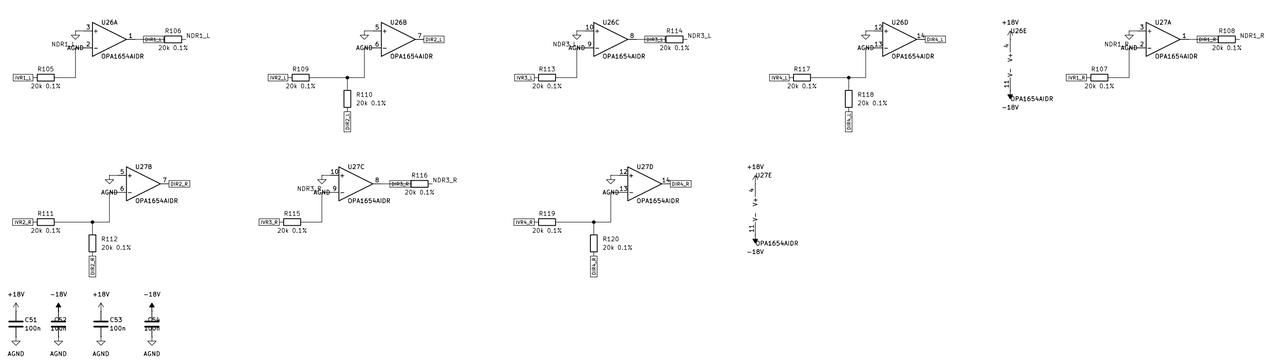
REROUTE to S3 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



REROUTE to S4 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



REROUTE INJECTION DRIVERS (2 OPA1654, inverting unity)



PWR_FLAG (for the ERC check only)



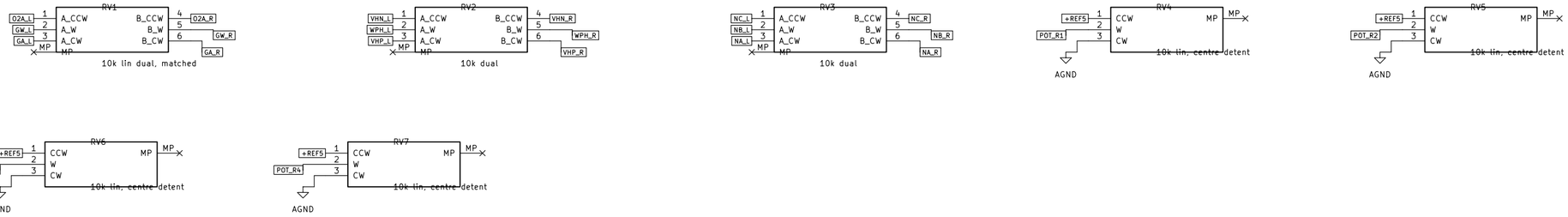
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand.

Sheet: /
File: fx-ret-reroute.kicad_sch

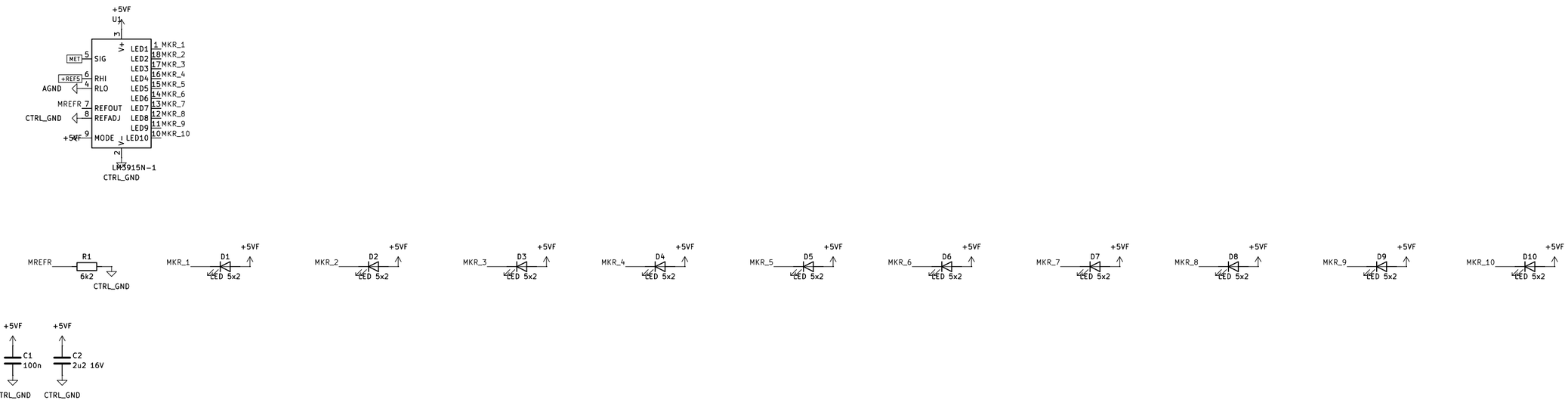
Title: maelcum FX-RET reroute (dec. 91) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

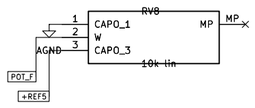
POTS: GAIN (dual), EQ2 (dual), REROUTE (single, centre detent)



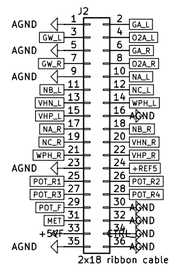
METER: one LM3915, 10 LEDs (+24…−3 dBu)



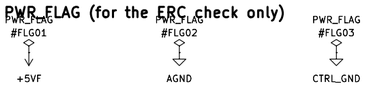
FADER (PTA4543, 45 mm travel, dec. 169)



CABLE: J2 to fx-ret.py (J2)



PWR_FLAG (for the ERC check only)



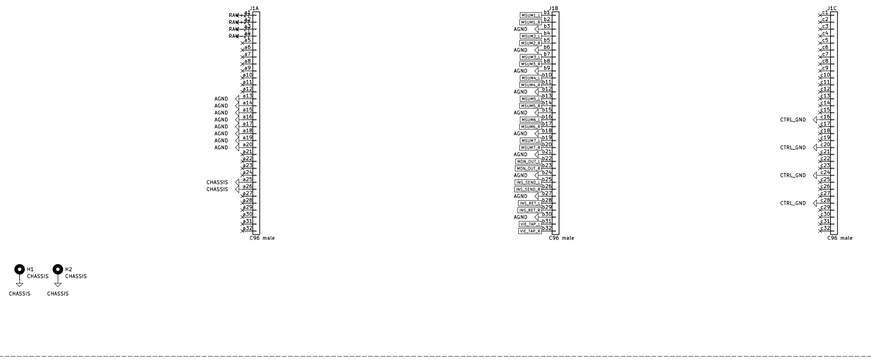
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: fx-ret-comandi.kicad_sch

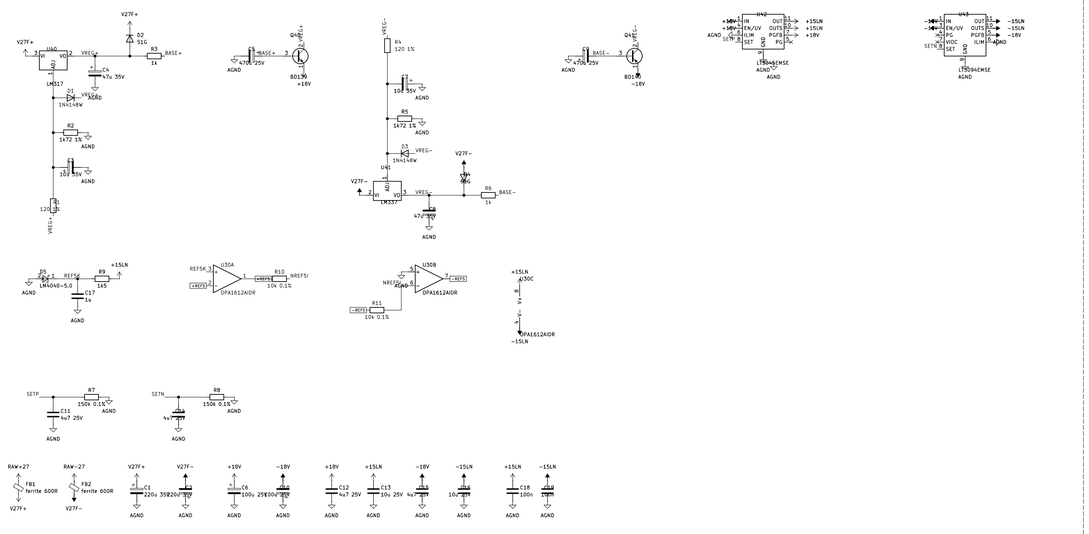
Title: maelcum FX-RET control board — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

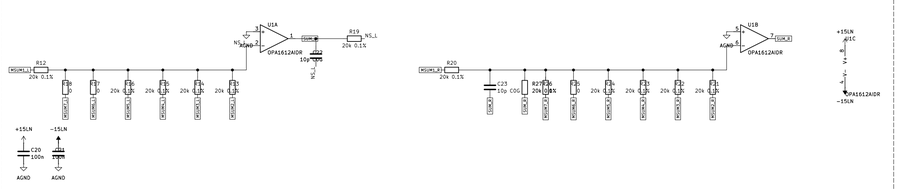
BACKPLANE C96 (MASTER class)



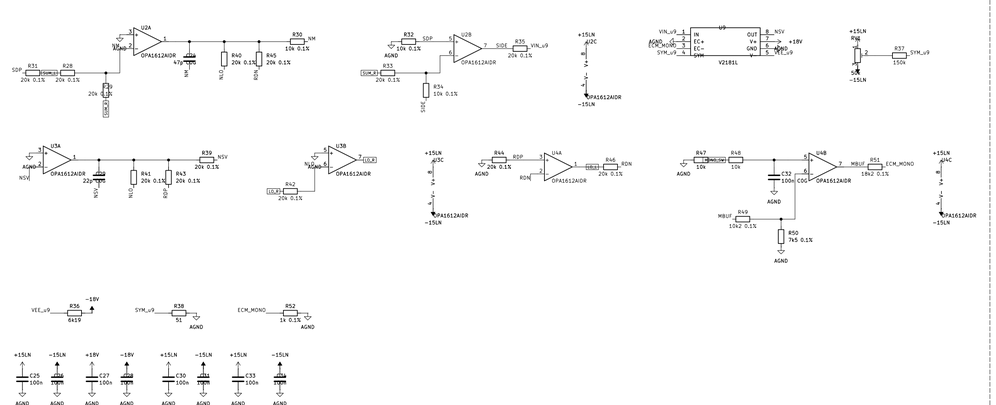
POWER +−18 (LM317/337 + capacitance multiplier), +−15 LN (LT3045/LT3094)



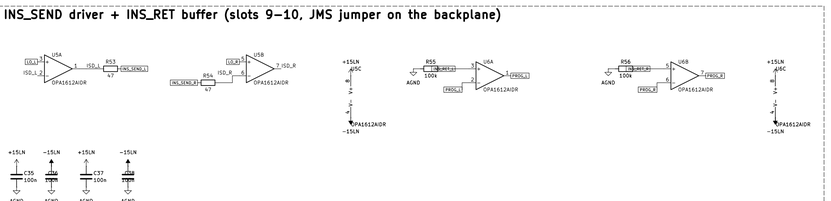
MASTER SUMMING [A2] (inverting, 20k/20k/10p, sum−noise−71A.cir)



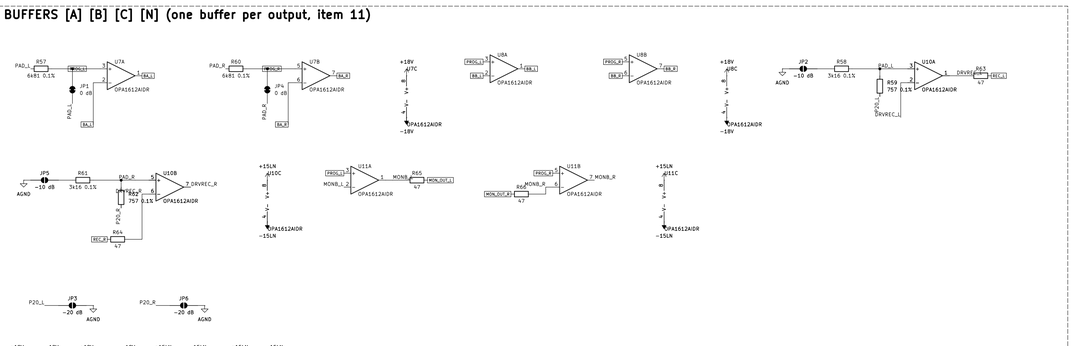
MONO [A3] (M/S matrix + VCA on S, mono−ms.cir; resistors NOTE)



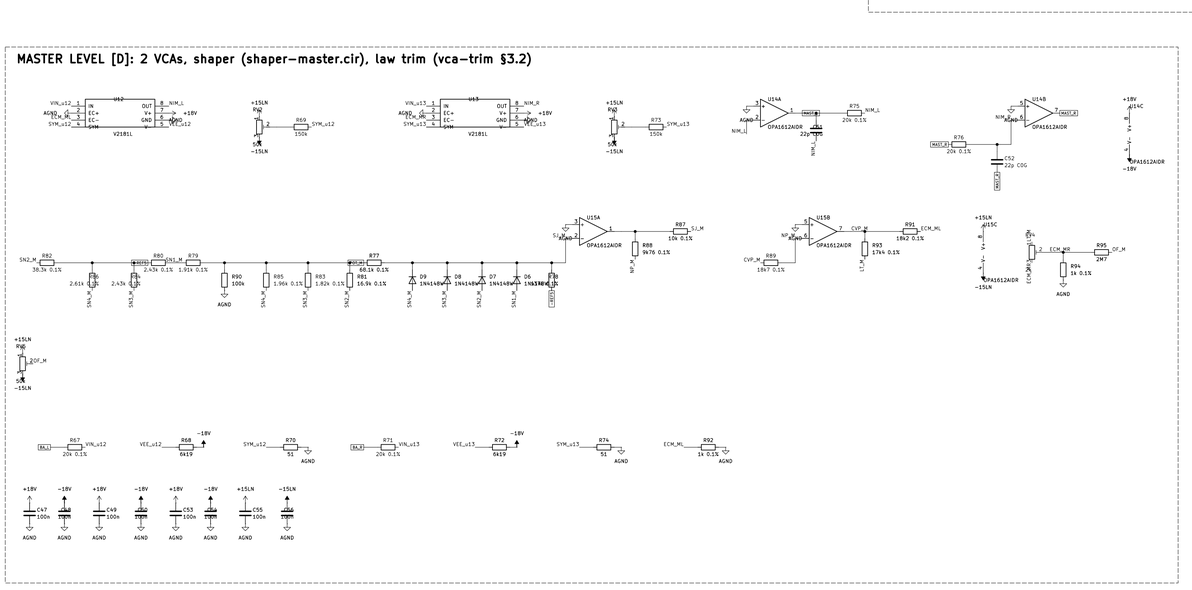
INS_SEND driver + INS_RET buffer (slots 9−10, JMS jumper on the backplane)



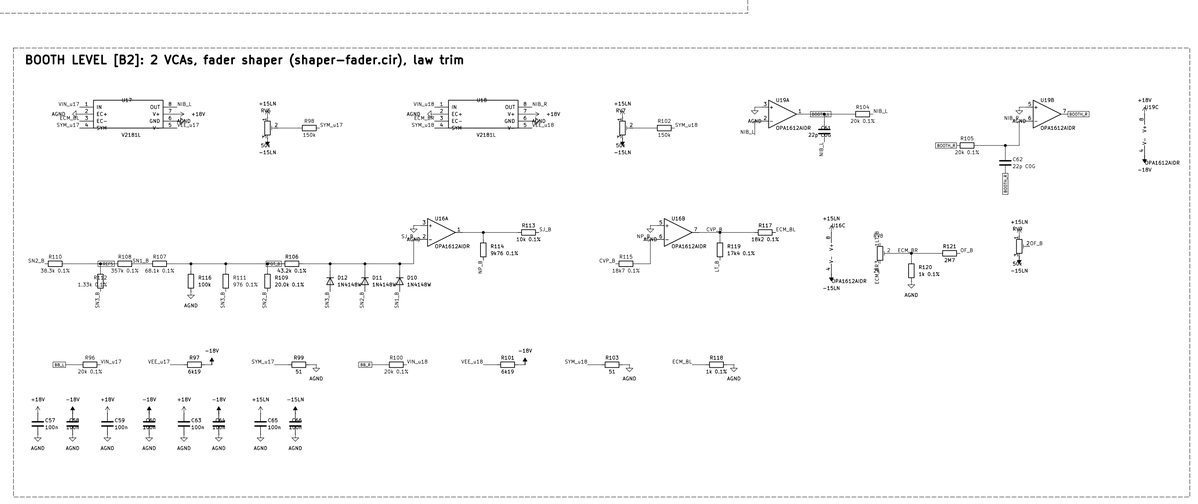
BUFFERS [A] [B] [C] [N] (one buffer per output, item 11)



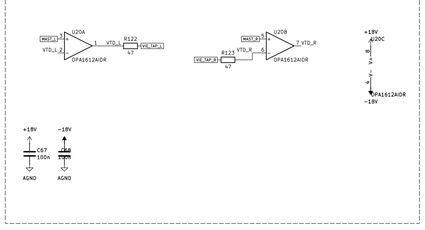
MASTER LEVEL [D]: 2 VCAs, shaper (shaper−master.cir), law trim (vca−trim §3.2)



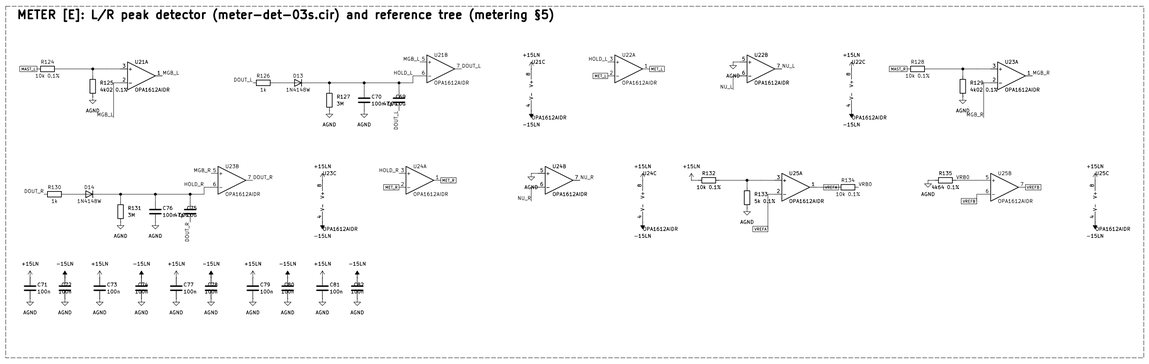
BOOTH LEVEL [B2]: 2 VCAs, fader shaper (shaper−fader.cir), law trim



VIE_TAP [V]: driver after MASTER LEVEL to b31−b32



METER [E]: L/R peak detector (meter−det−03s.cir) and reference tree (metering §5)



CONNECTORS: J2 to the OUTPUT board, J3 to the CONTROL board



PWR_FLAG (for the ERC check only)



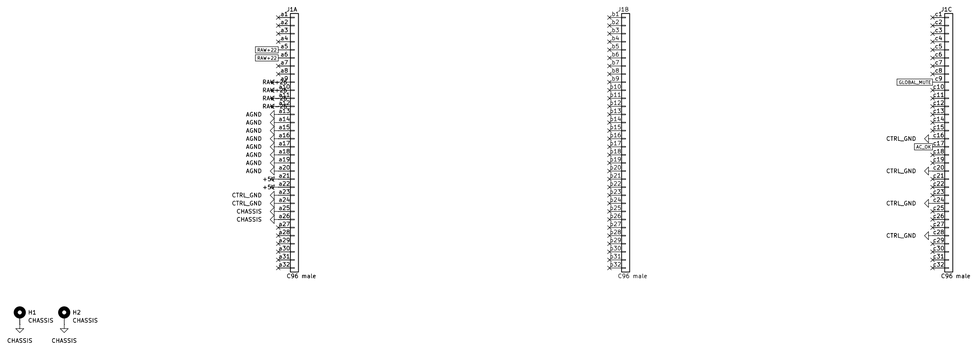
Generated by pcb/tools/legglbile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: master−segnale.kicad_sch

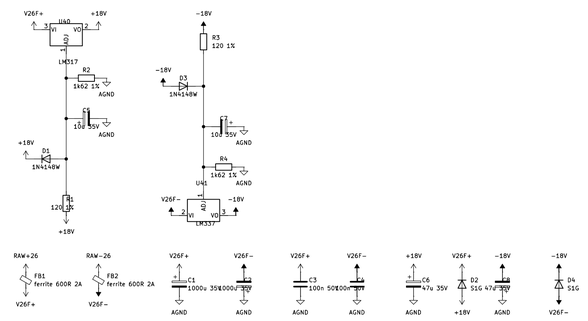
Title: maelcum MASTER signal board (dec. 140) − draft v0.1 − readable sch

Size: User	Date:
KiCad E.D.A. 10.0.7	Rev: 0.2−draft
	Id: 1/1

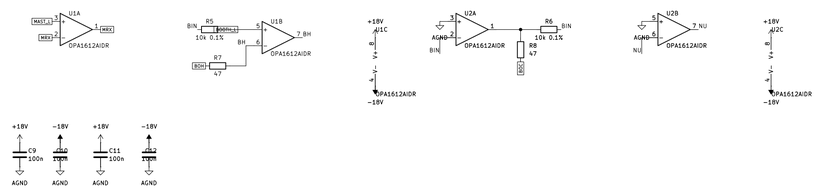
BACKPLANE C96 (MASTER-OUT class, dec. 143): power only



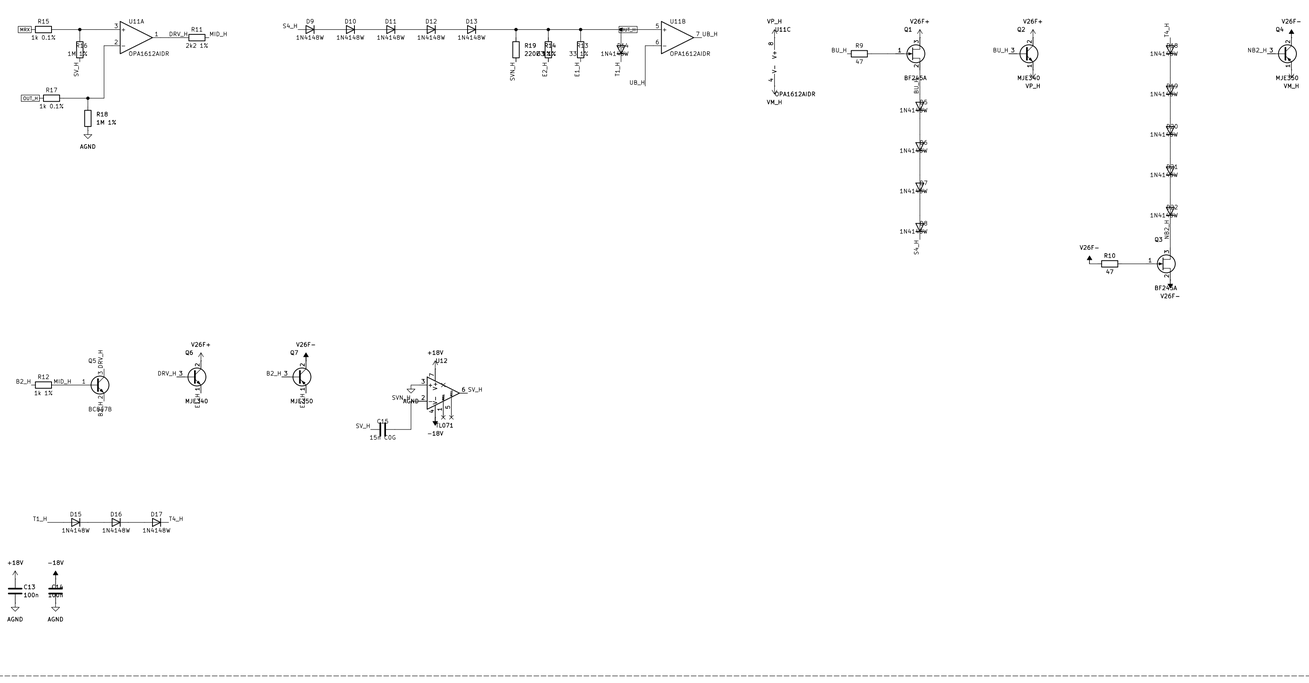
POWER [M]: filtered +-26 static rails, local +-18 (LM317/337)



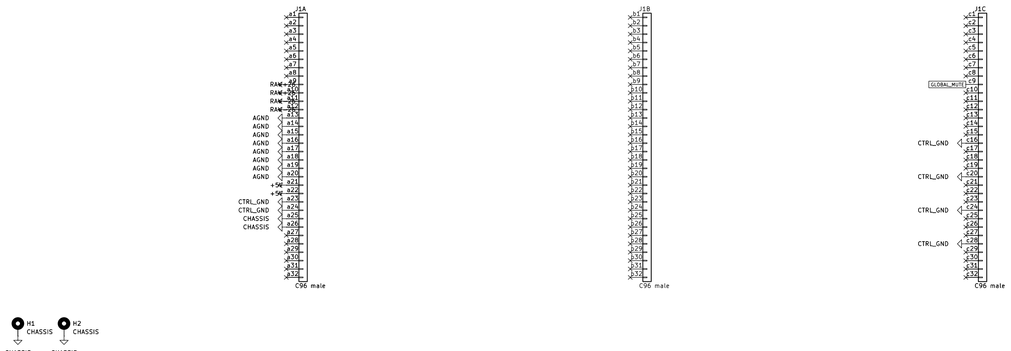
MASTER RECEIVER and BOOTH DRIVER [B3] (direct OPA1612, §3.4)



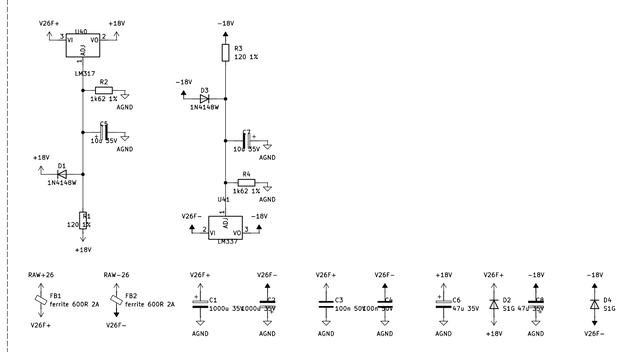
OUTPUT STAGE hot [F]+[G] (bootstrap, dec. 86)



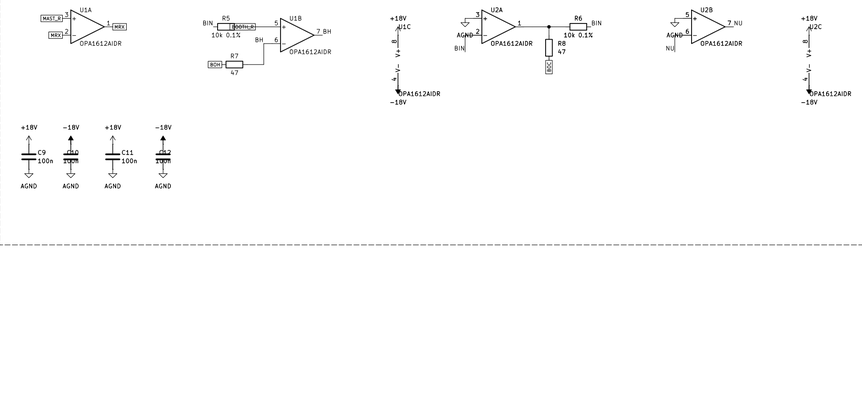
BACKPLANE C96 (MASTER-OUT class, dec. 143): power only



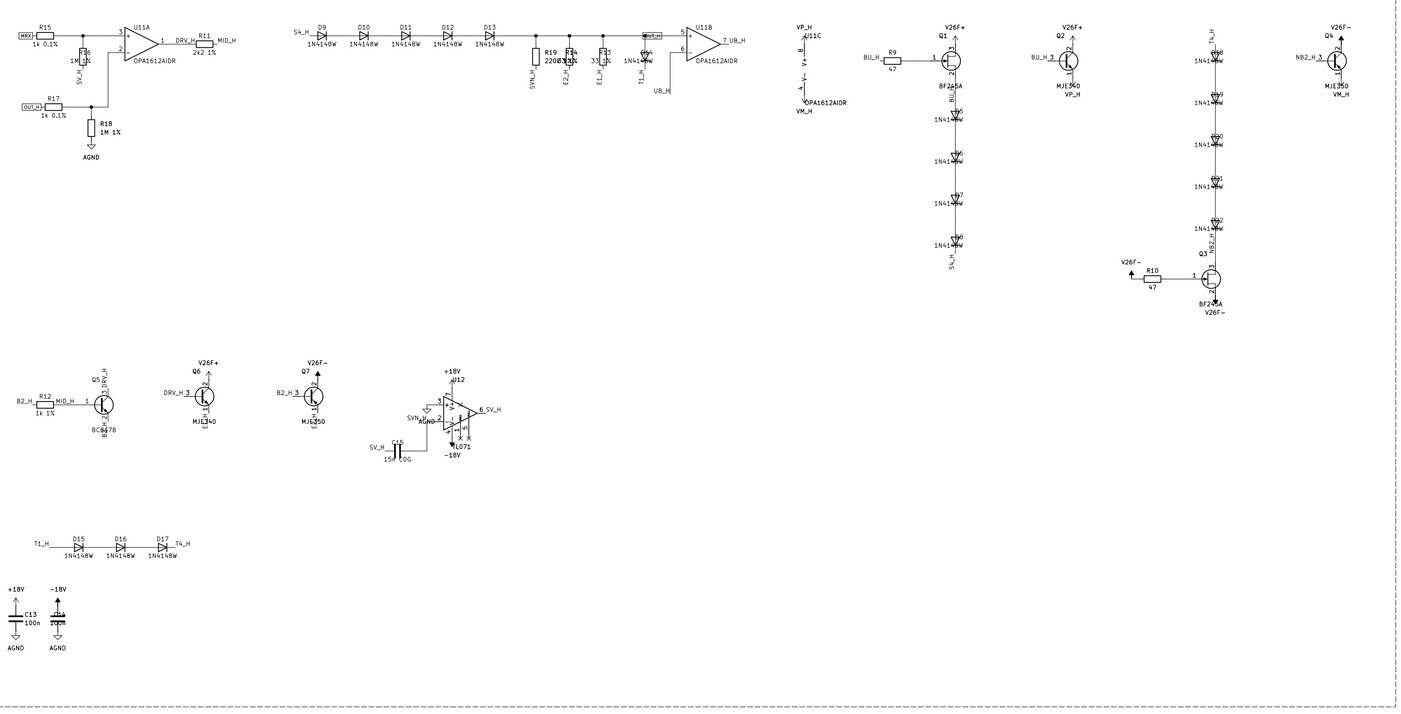
POWER [M]: filtered +-26 static rails, local +-18 (LM317/337)



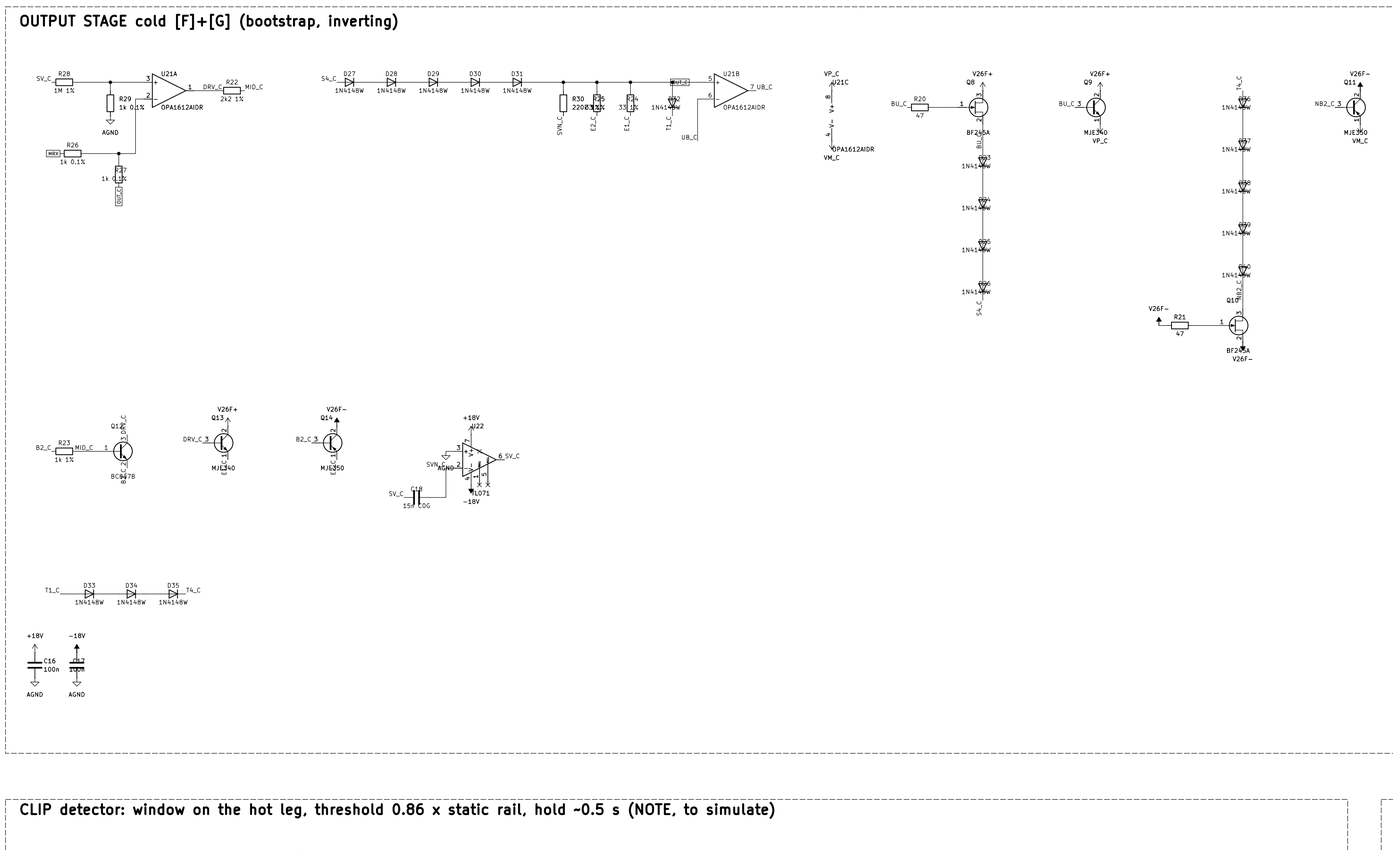
MASTER RECEIVER and BOOTH DRIVER [B3] (direct OPA1612, §3.4)



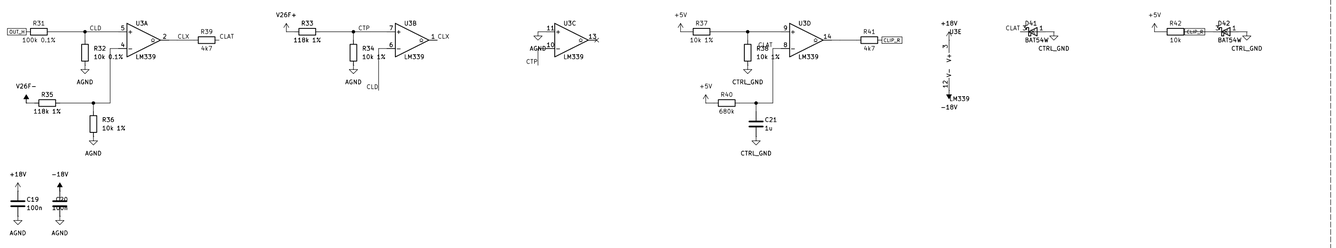
OUTPUT STAGE hot [F]+[G] (bootstrap, dec. 86)



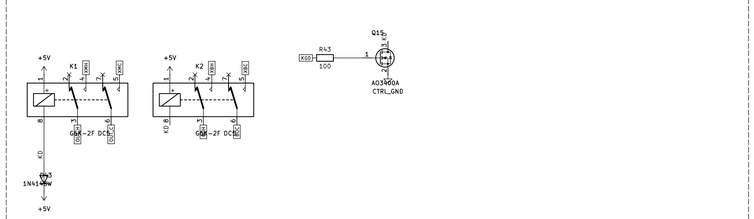
OUTPUT STAGE cold [F]+[G] (bootstrap, inverting)



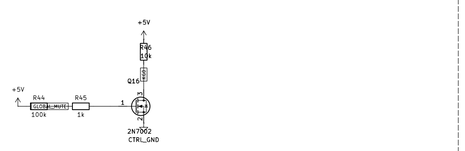
CLIP detector: window on the hot leg, threshold 0.86 x static rail, hold ~0.5 s (NOTE, to simulate)



MASTER and BOOTH RELAYS (G6K-2F, 5 V, 237 ohm; energised = sound, failsafe)



RELAY DRIVE from GLOBAL_MUTE (c9): low = sound



CONNECTORS: J2 from the signal board, J3 to the rear connector board



PWR_FLAG (for the ERC check only)



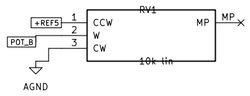
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

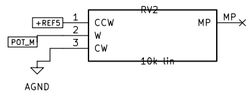
Sheet: /
File: master-uscita-R.kicad_sch

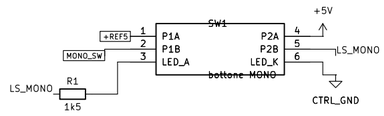
Title: maelcum MASTER OUTPUT board R (dec. 142-143) - draft v0.1 - rea

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

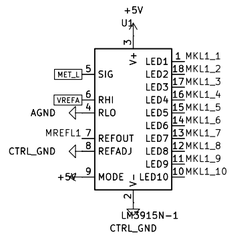
POTS and MONO BUTTON (positions from the panel render)

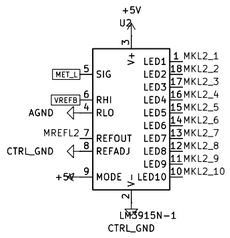


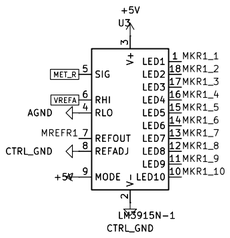


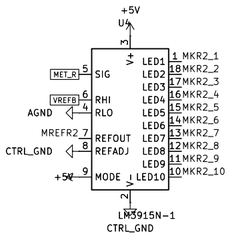


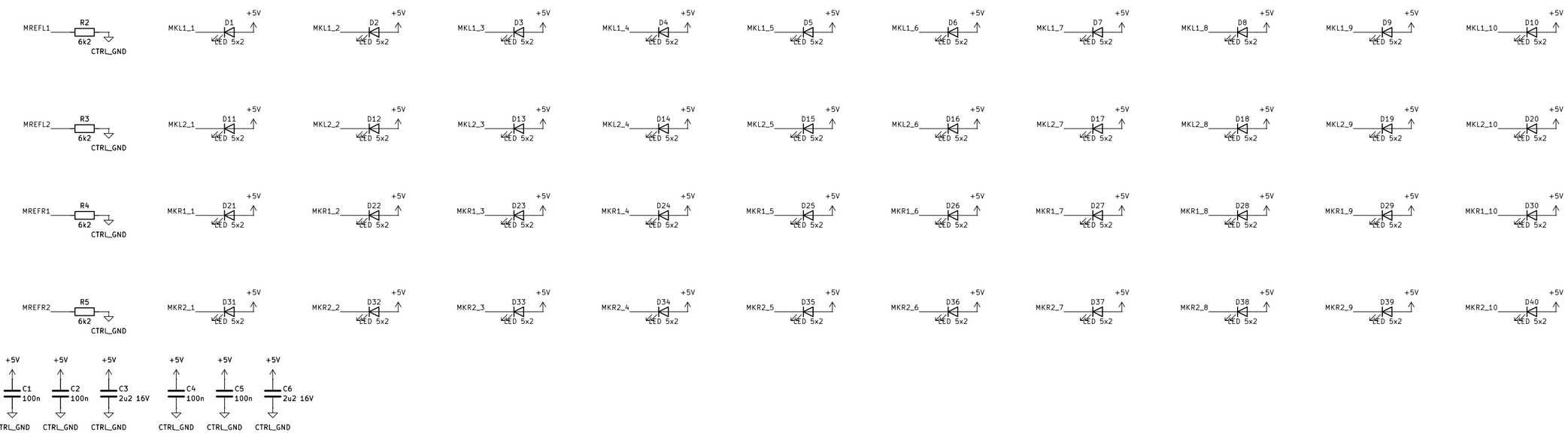
METERS L and R: 2 cascaded LM3915 per side, 20 LEDs



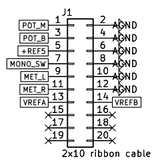


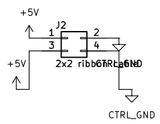




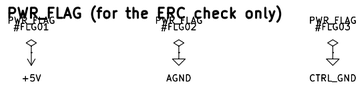


CABLES: J1 to the signal board (J3), J2 to the L output board (J4)





PWR_FLAG (for the ERC check only)



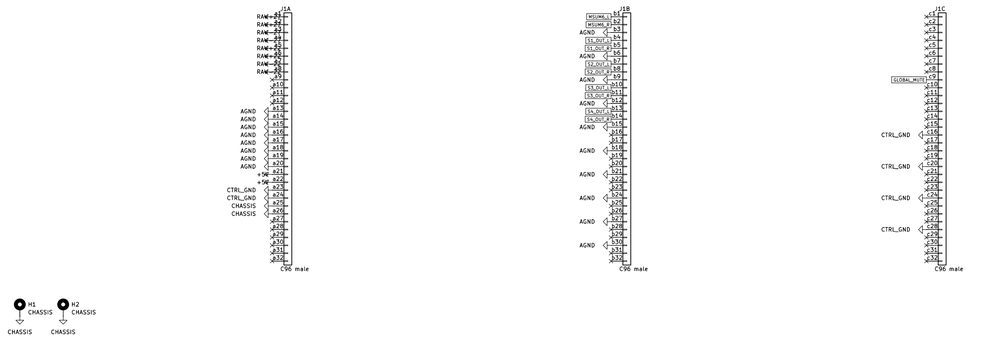
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: master-comandi.kicad_sch

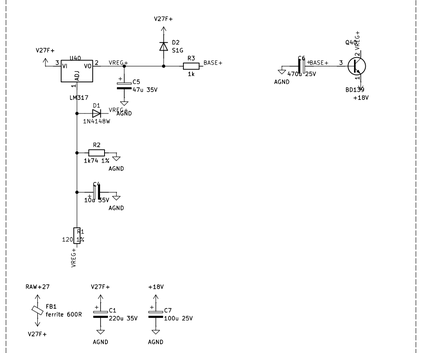
Title: maelcum MASTER control board — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

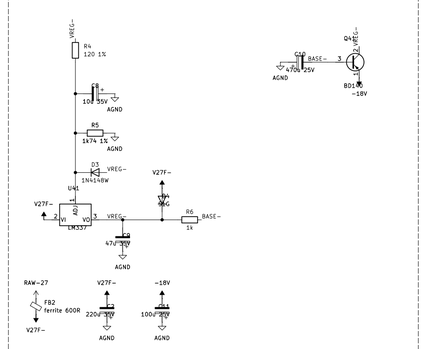
BACKPLANE C96 (MIC/GEN class, MSUM6)



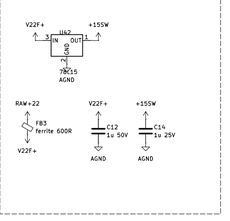
POWER: +18V (LM317 + BD139)



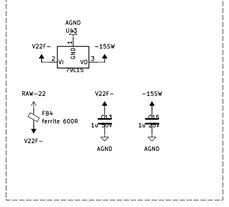
POWER: −18V (LM337 + BD140)



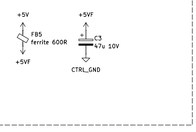
POWER: +15SW (78L15)



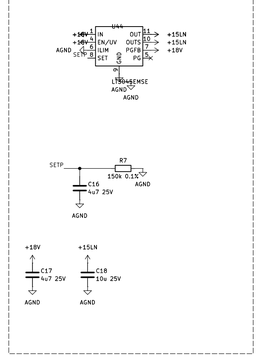
POWER: −15SW (79L15)



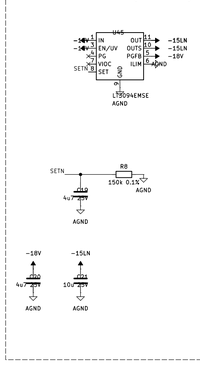
POWER: +5VF



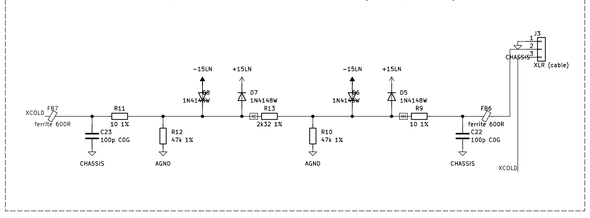
POWER: +15LN (LT3045)



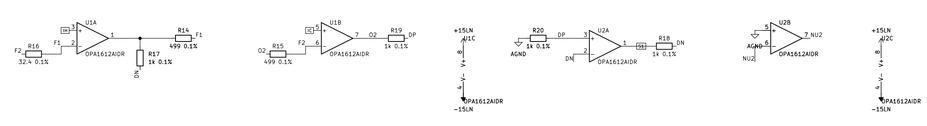
POWER: −15LN (LT3094)



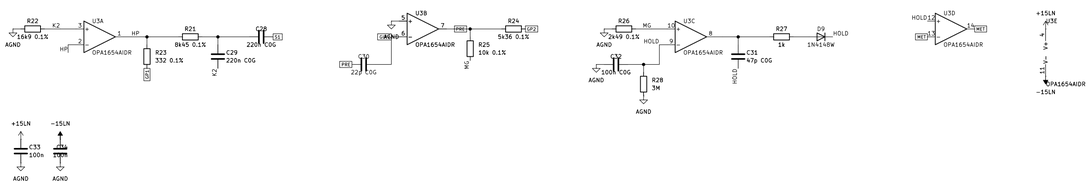
XLR INPUT: RFI, clamp, 2.32k load, 2x47k bias (mic−pre.inc)



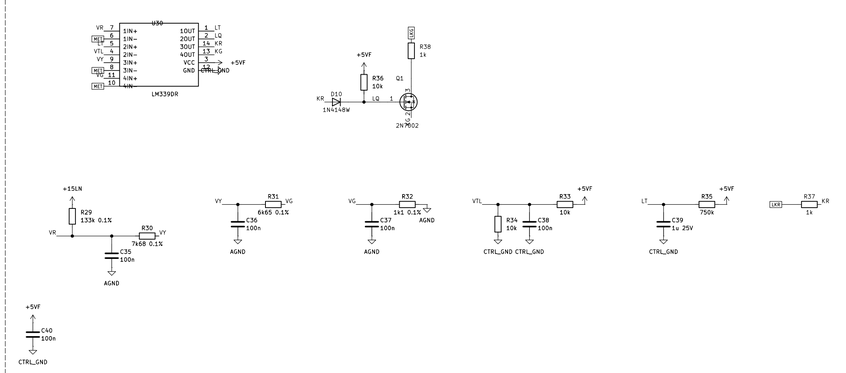
INA +30.05 dB and unity differential (OPA1612)



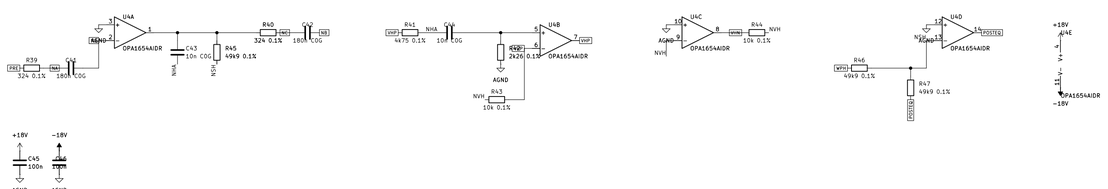
HIGH−PASS 60 Hz, STAGE 2 (0...+30 dB), LED DETECTOR (OPA1654)



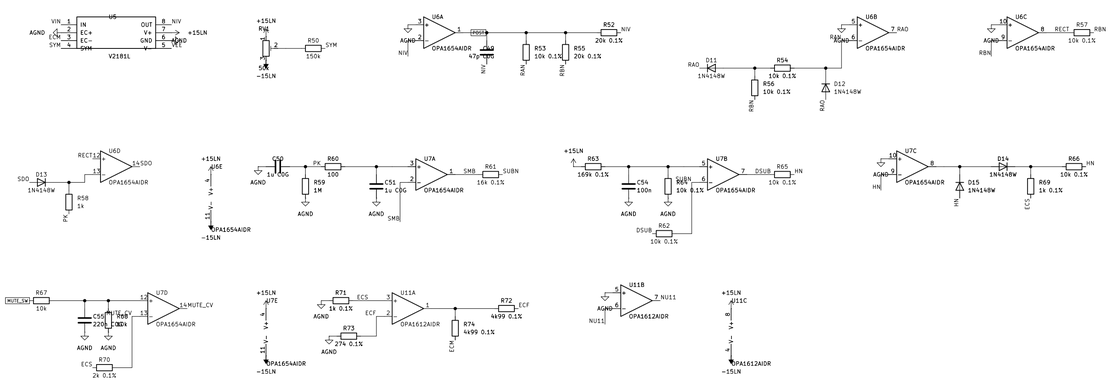
Three−colour LED (LM339)



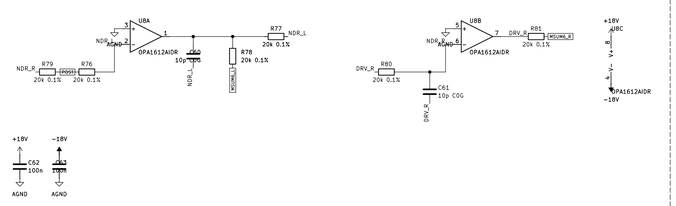
EQ2 (+−18)



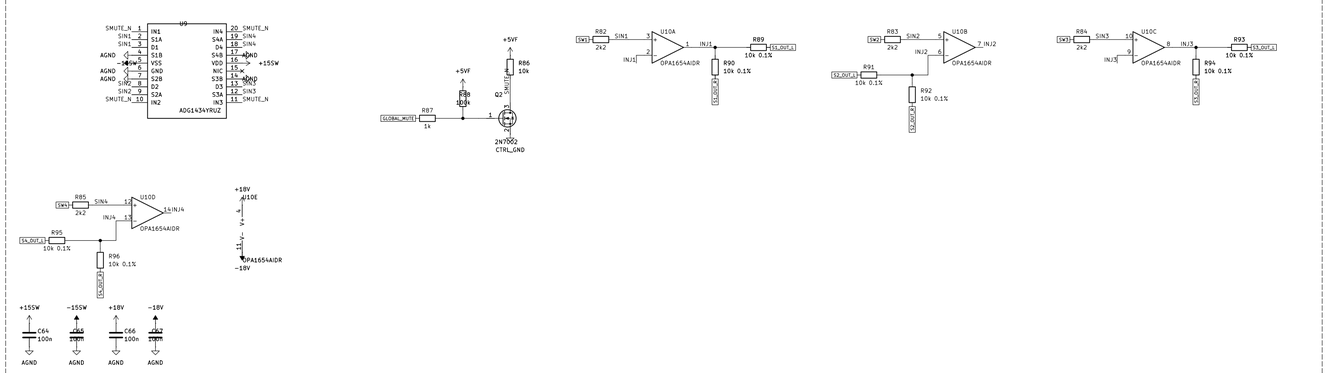
MUTE and limiter VCA (dec. 111, 145; sim/mic/mic−limiter.cir)



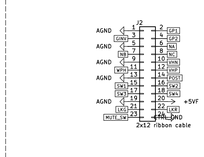
MSUM6 L/R DRIVERS (inverting, todo 227)



SENDS S1−S4 (pots on the cable, global mute, injection)



CONTROL BOARD CABLE (J2)



PWR_FLAG (for the ERC check only)



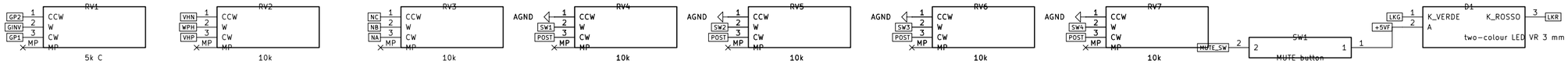
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mic.kicad_sch

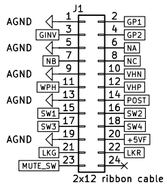
Title: maelcum MIC (dec. 97) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1−draft
KiCad E.D.A. 10.0.7		Id: 1/1

CONTROLS (positions from the panel render)



CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)

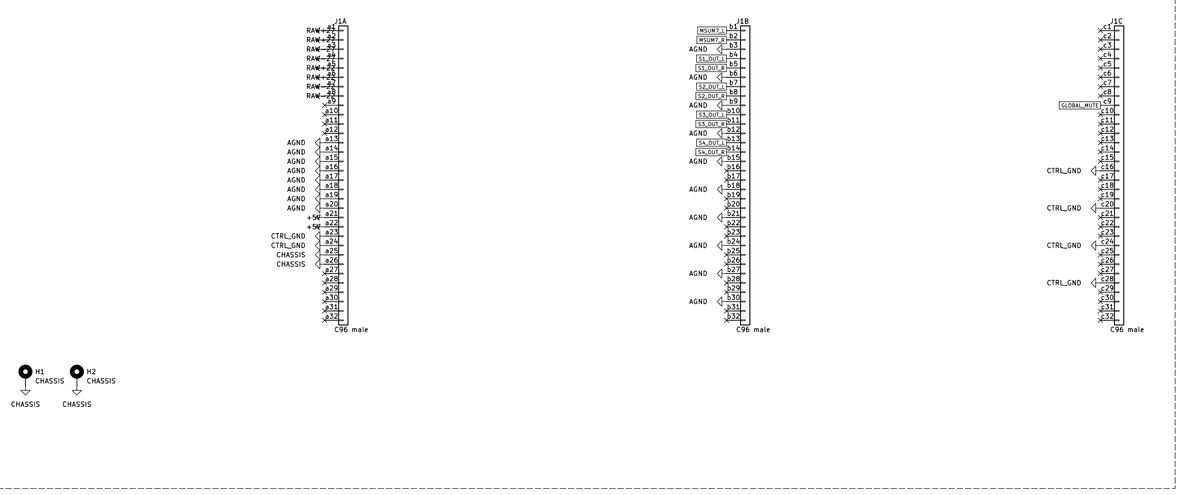
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mic-comandi.kicad_sch

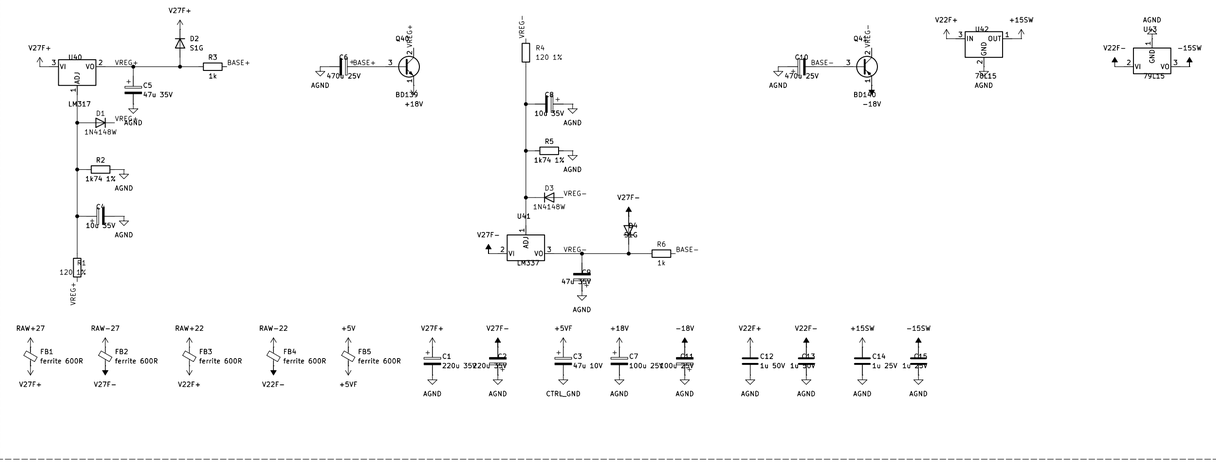
Title: maelcum MIC control board — draft v0.1 — readable schematic

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

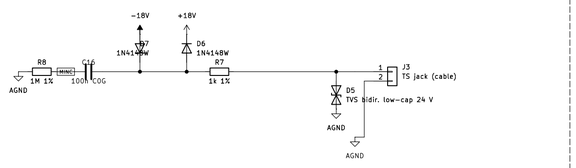
BACKPLANE C96 (MIC/GEN class, MSUM7)



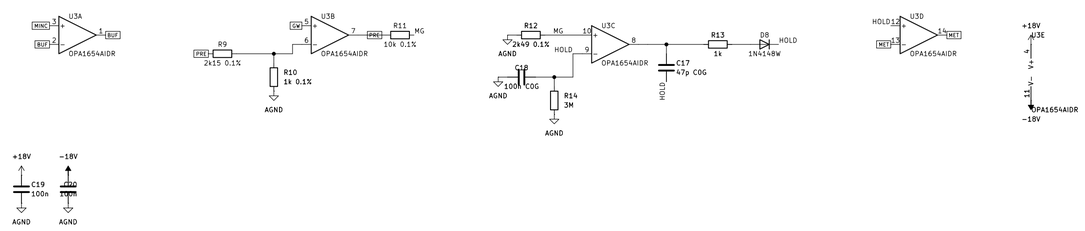
POWER +−18 audio, +−15 switch, +5



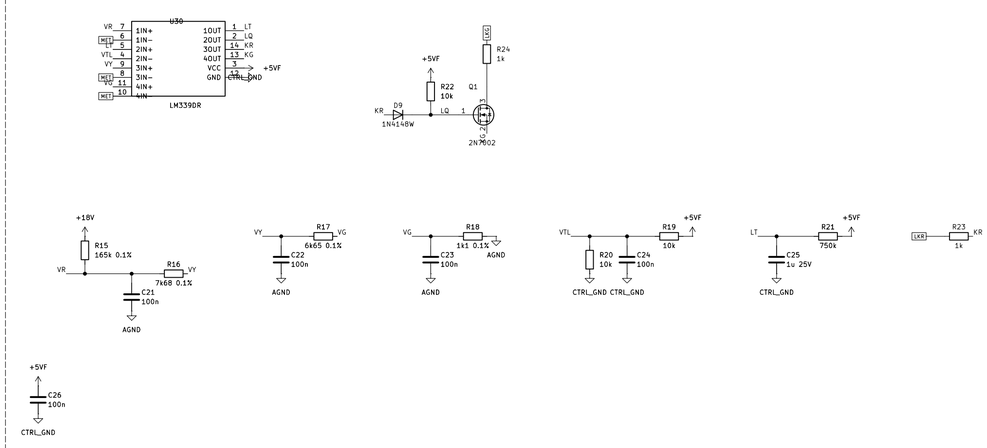
TS JACK INPUT: series R, clamp, TVS, film C, 1M bias



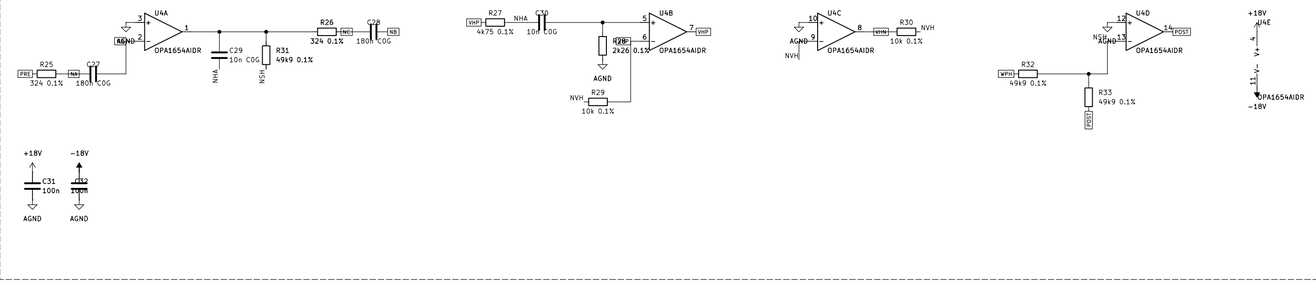
BUFFER, MAKE−UP +10 dB, LED DETECTOR (OPA1654)



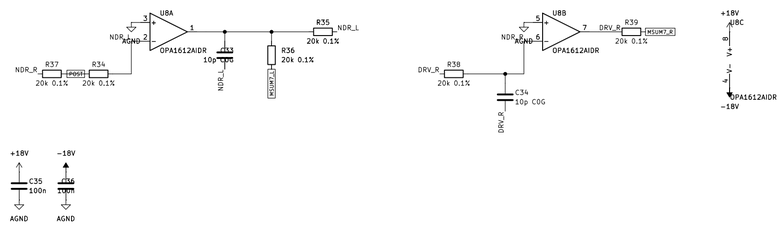
Three−colour LED (LM339)



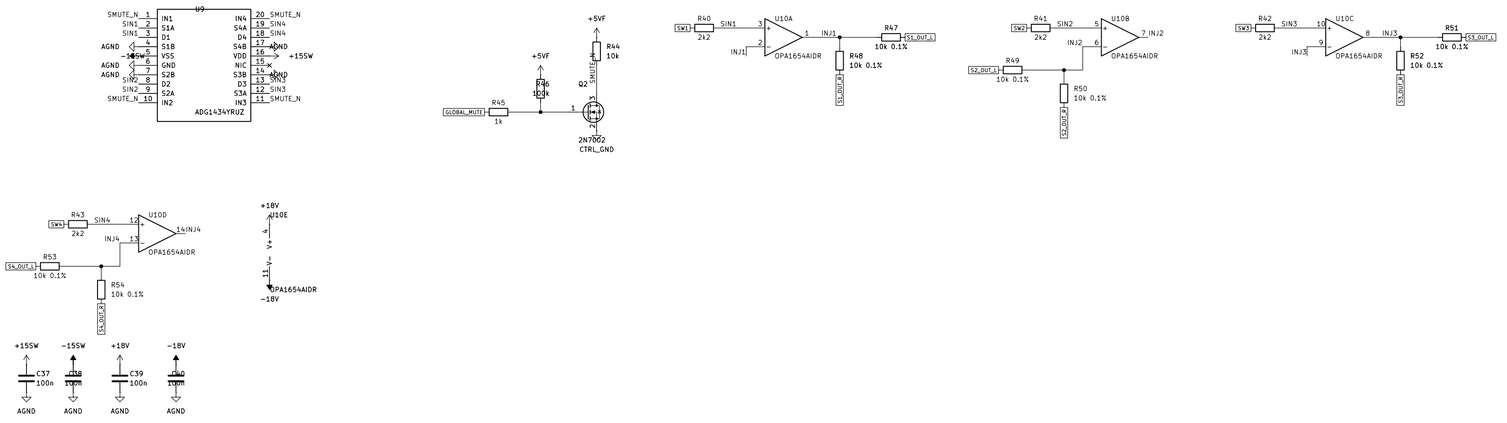
EQ2 (+−18)



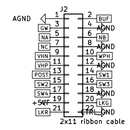
MSUM7 L/R DRIVERS (inverting, todo 227)



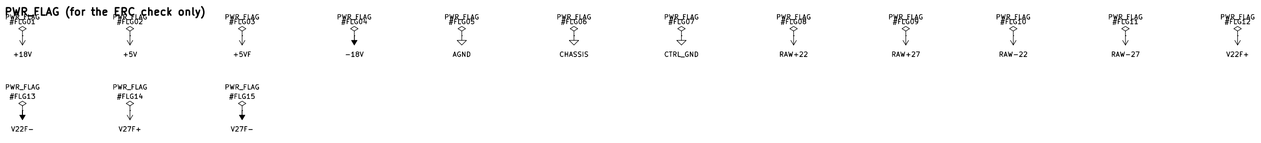
SENDS S1−S4 (pots on the cable, global mute, injection)



CONTROL BOARD CABLE (J2)



PWR_FLAG (for the ERC check only)



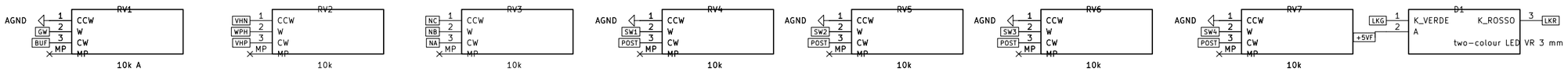
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mono.kicad_sch

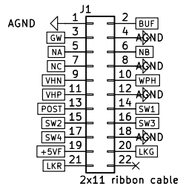
Title: maelcum MONO (dec. 97) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1−draft
KiCad E.D.A. 10.0.7		Id: 1/1

CONTROLS (positions from the panel render)



CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)



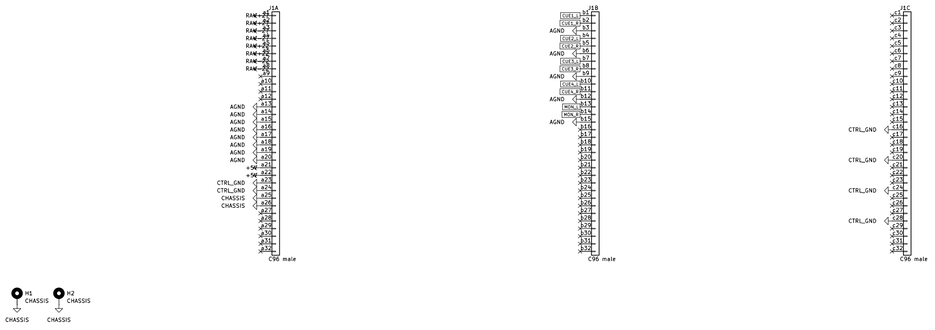
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mono-comandi.kicad_sch

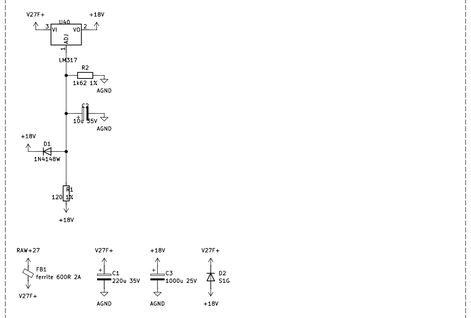
Title: maelcum MONO control board — draft v0.1 — readable schematic

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

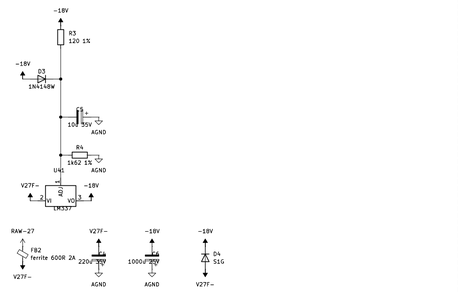
BACKPLANE C96 (CUE class: slots 1–4 = CUE1–4 in, slot 5 = MON in)



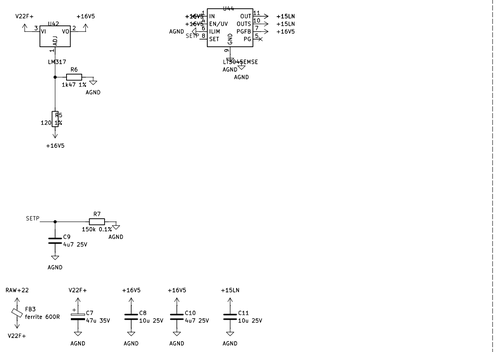
POWER +18 audio (LM317 from raw +27, 1000 uF bulk)



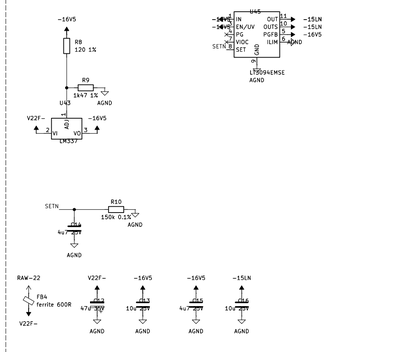
POWER −18 audio (LM337 from raw −27, 1000 uF bulk)



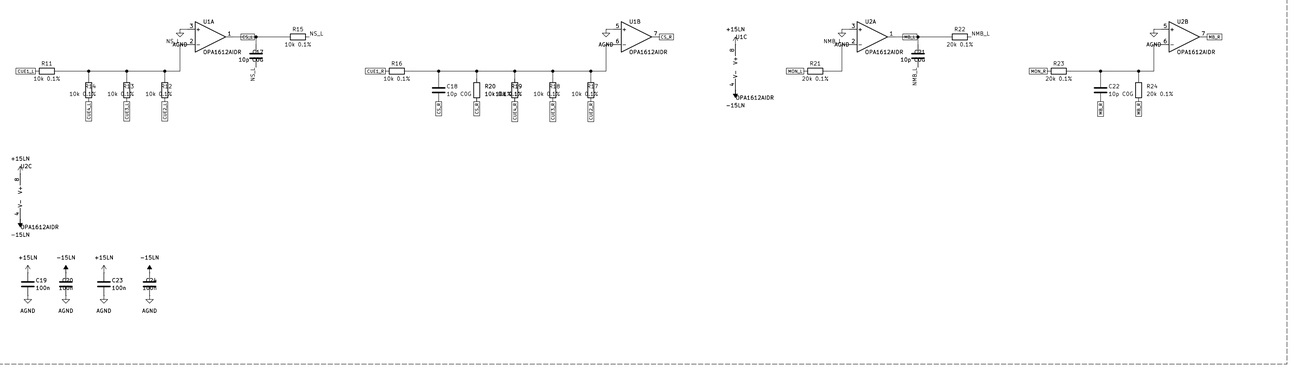
POWER +15 LN (LM317 to +16.5 from raw +22, then LT3045)



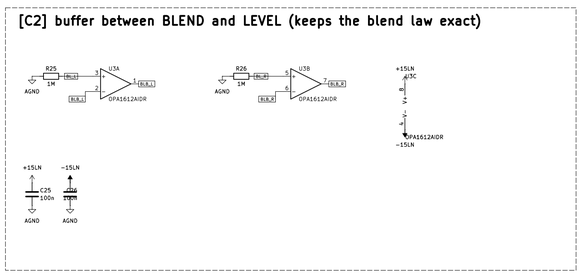
POWER −15 LN (LM337 to −16.5 from raw −22, then LT3094)



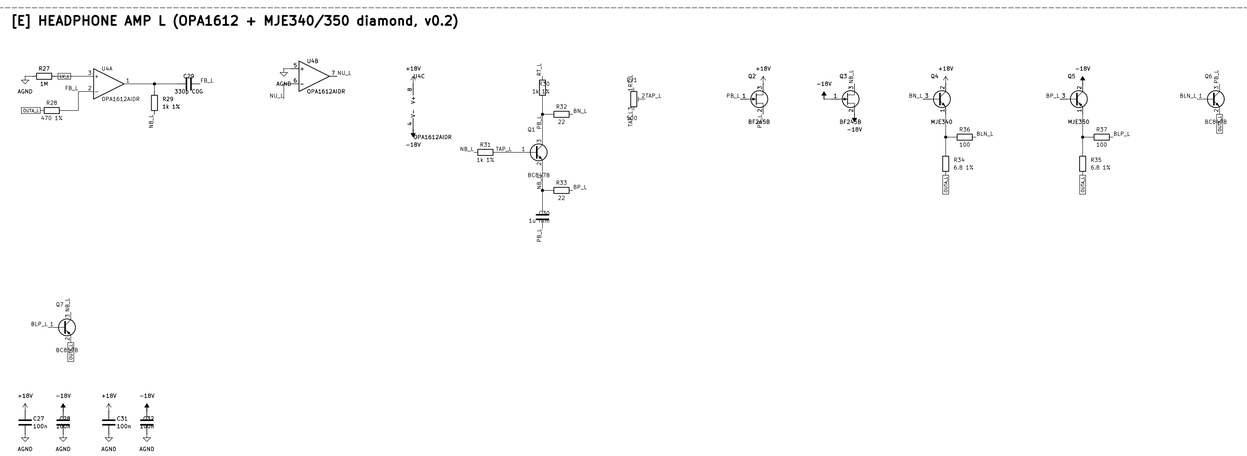
[A] CUE SUMMING L/R (virtual earth, Rf 10k, −1) and [B] MON (inverting 20k/20k, F4)



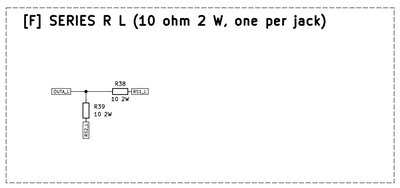
[C2] buffer between BLEND and LEVEL (keeps the blend law exact)



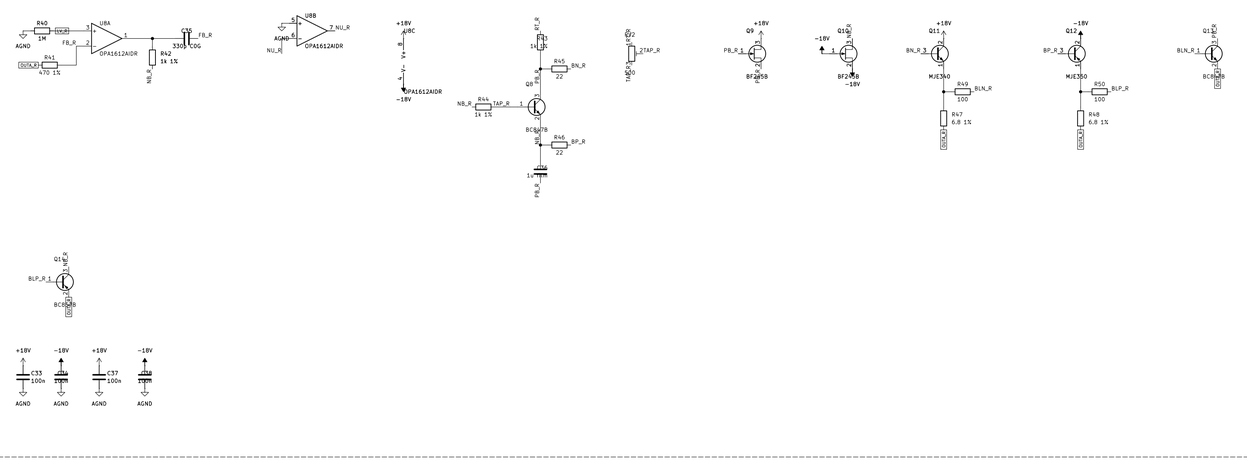
[E] HEADPHONE AMP L (OPA1612 + MJE340/350 diamond, v0.2)



[F] SERIES R L (10 ohm 2 W, one per jack)



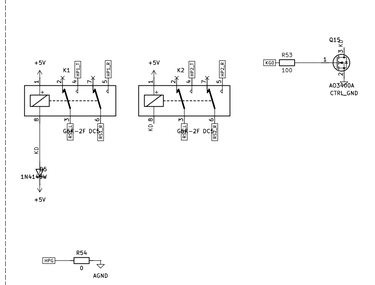
[E] HEADPHONE AMP R (OPA1612 + MJE340/350 diamond, v0.2)



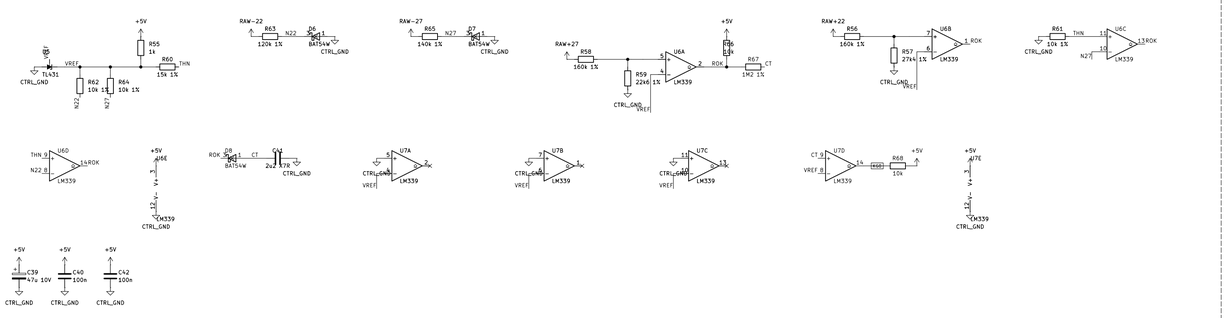
[F] SERIES R R (10 ohm 2 W, one per jack)



[F] HEADPHONE RELAYS (G6K−2F, 5 V, 237 ohm; energised = sound; failsafe)



[G] PROTECTION: windows on raw +−22 and +−27, 1.8 s timer (the CUE ignores c9)



CABLES to the control board: J2 pots, J3 headphones



PWR_FLAG (for the ERC check only)



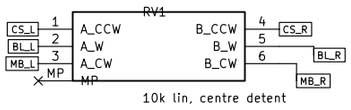
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

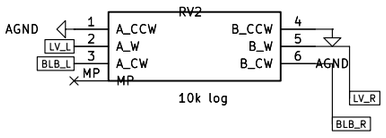
Sheet: /
File: cue.kicad_sch

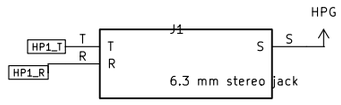
Title: maelcum CUE / cuffie (sistema-cue v0.2) — draft v0.1 — readable sc

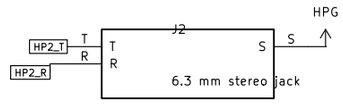
Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

BLEND and LEVEL POTS, HEADPHONE JACKS (positions from the panel render)

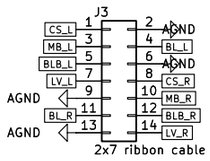


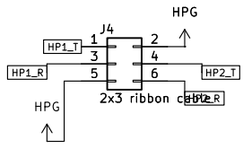




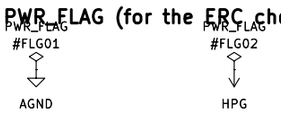


CABLES: J3 pots (J2 of cue.py), J4 headphones (J3 of cue.py)





PWR_FLAG (for the ERC check only)



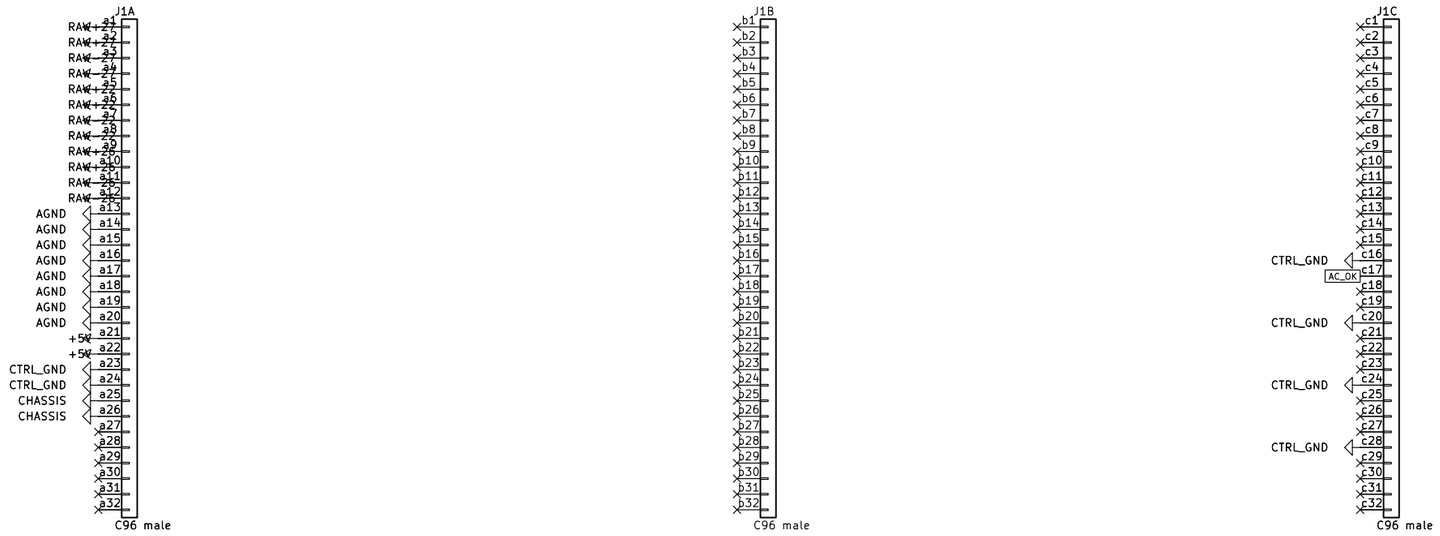
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: cue-comandi.kicad_sch

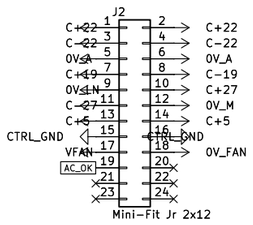
Title: maelcum CUE control board — draft v0.1 — readable schematic

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

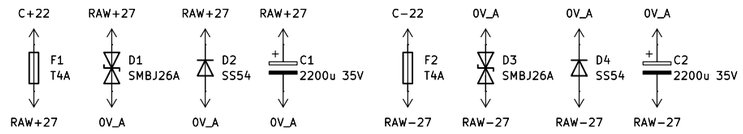
BACKPLANE C96 (PSU-IN slot): whole row a, AC_OK on c17



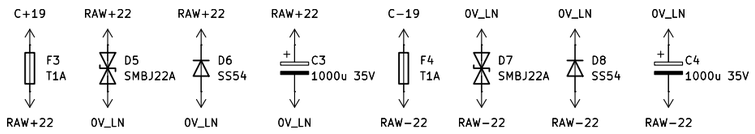
CABLE FROM THE BOX (20 conductors, §3)



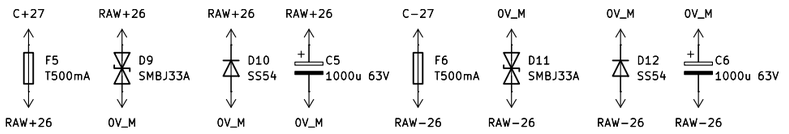
AUDIO RAIL +-22.0 V (raw +-27 pins): T4 A, TVS 26 V, 2.2 mF



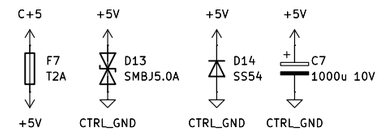
LN RAIL +-19 V (raw +-22 pins): T1 A, TVS 22 V, 1 mF



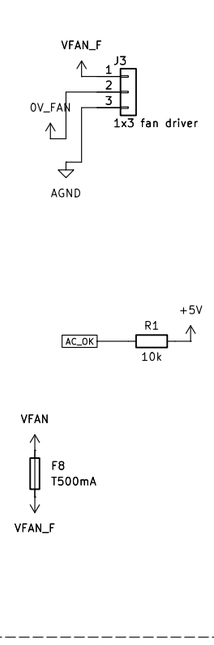
RAIL [M] raw +-27 V (raw +-26 pins): T500 mA, TVS 33 V, 1 mF



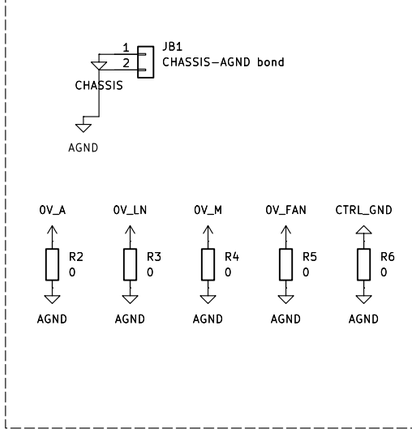
RAIL +5 V: T2 A, TVS 5.0 V, 1 mF



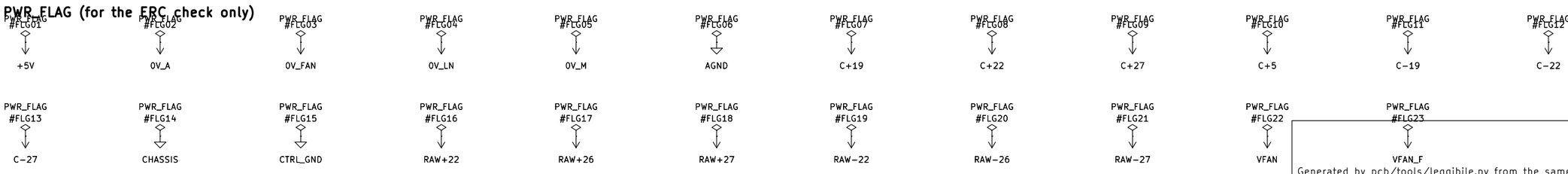
V_FAN (T500 mA) and AC_OK



analogue-0V STAR (§4.2, grounding §2)



PWR_FLAG (for the ERC check only)



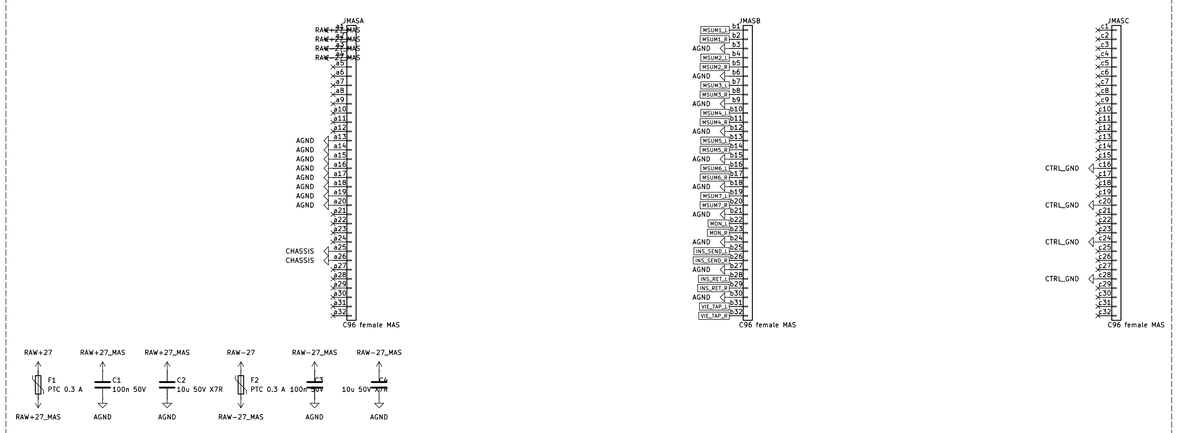
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: psu-in.kicad_sch

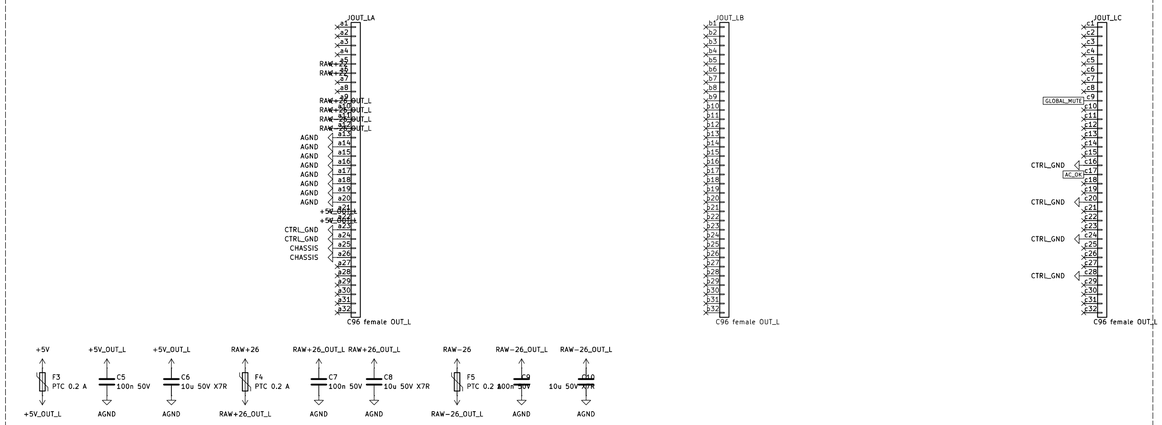
Title: maelcum PSU-IN (dec. 113) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

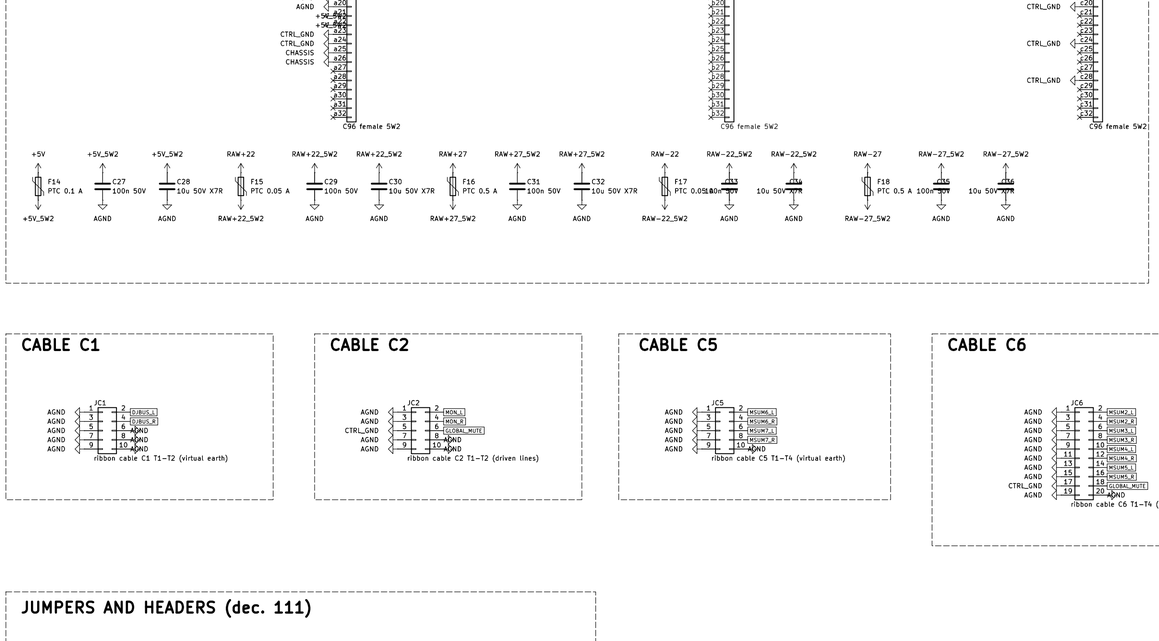
SLOT MAS (MASTER class)



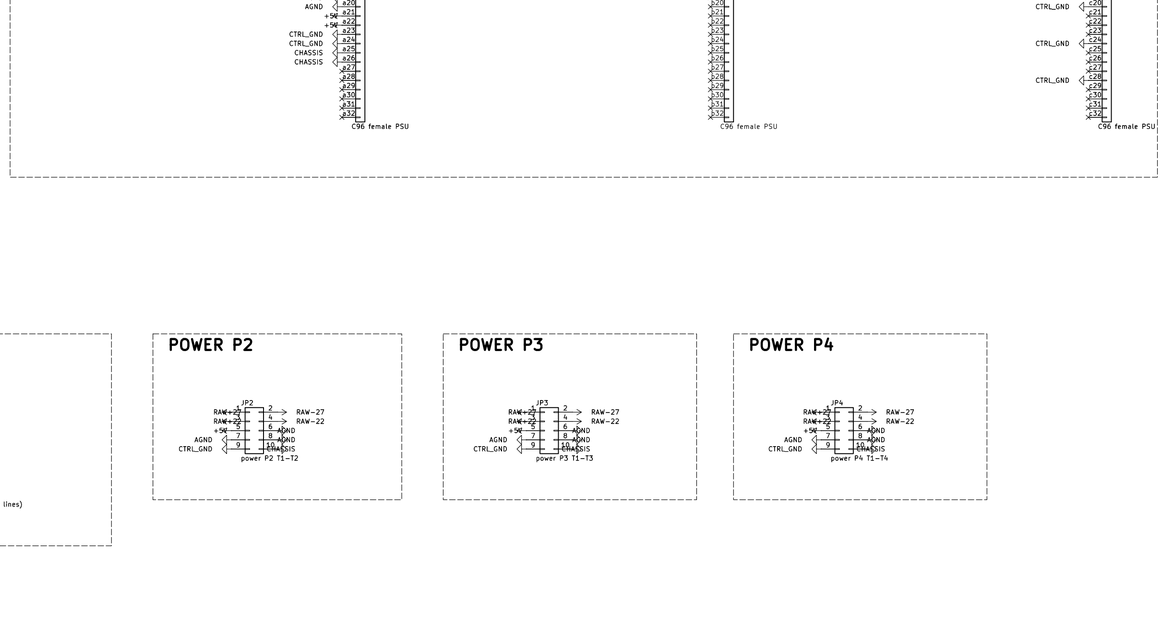
SLOT OUT_L (MASTER-OUT class)



SLOT OUT_R (MASTER-OUT class)



SLOT 5W1 (5-WAY class)



SLOT 5W2 (5-WAY class)

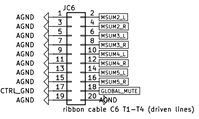
SLOT PSU (PSU-IN class)

CABLE C1

CABLE C2

CABLE C5

CABLE C6

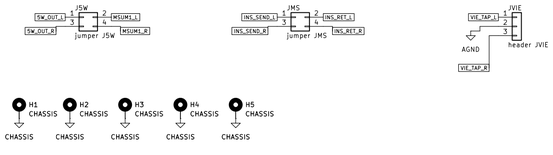


POWER P2

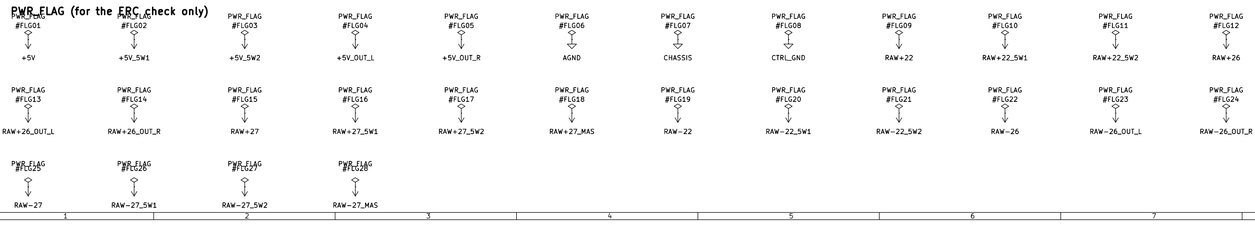
POWER P3

POWER P4

JUMPERS AND HEADERS (dec. 111)



PWR_FLAG (for the ERC check only)



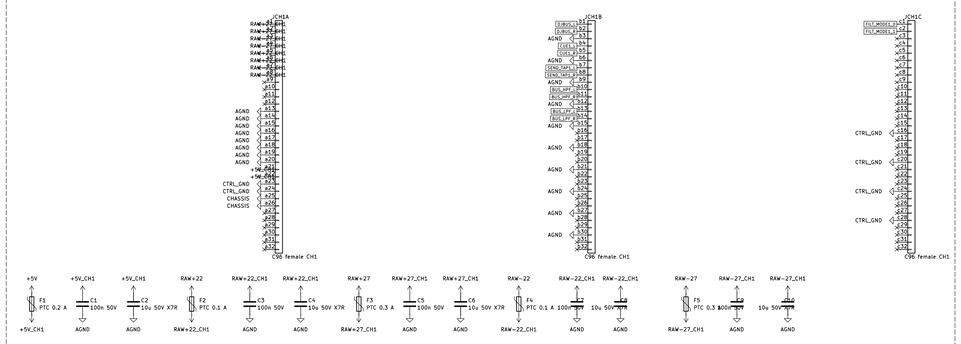
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: backplane-t1.kicad_sch

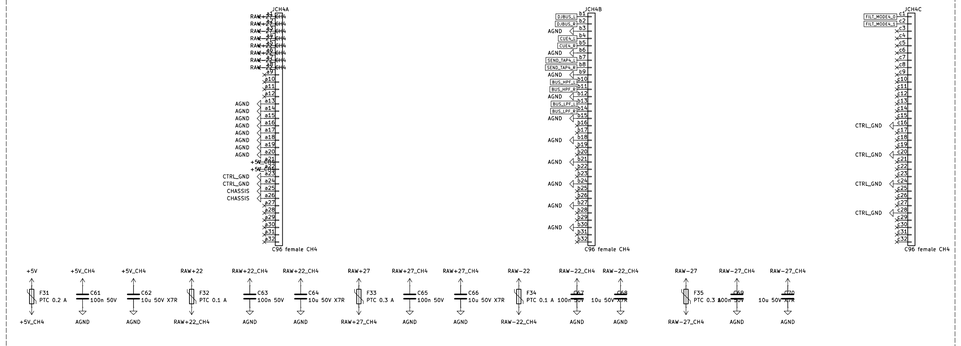
Title: maelcum backplane T1 (rear: MASTER, 5-WAY, PSU-IN) – draft v0.1 –

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

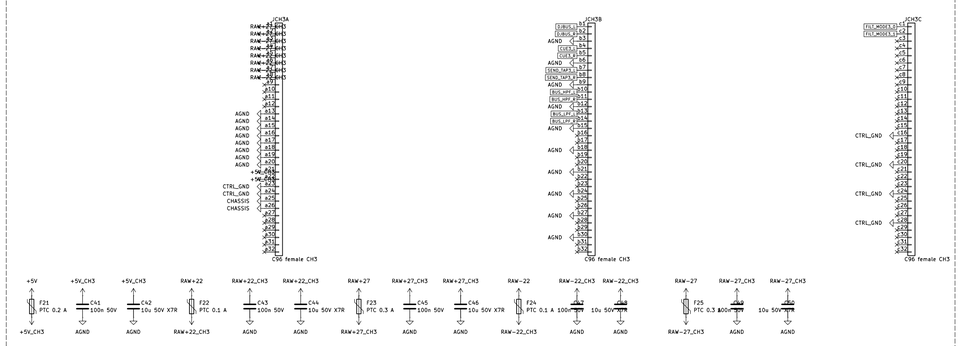
SLOT CH1 (CHANNEL class)



SLOT CH2 (CHANNEL class)

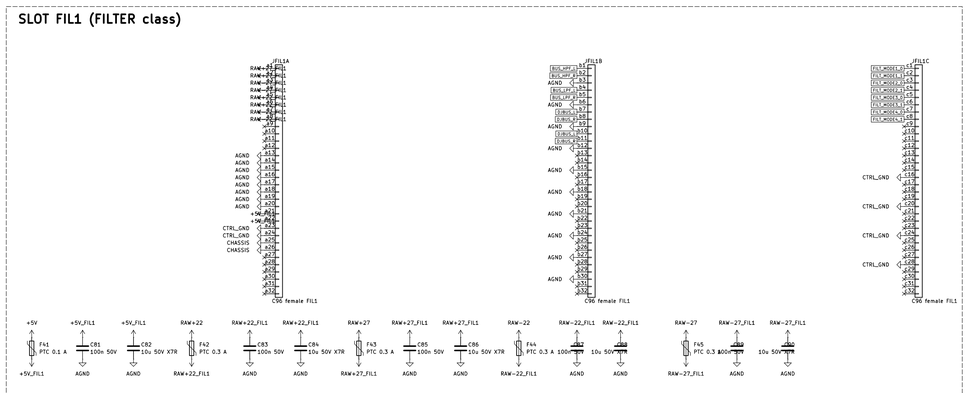


SLOT CH3 (CHANNEL class)

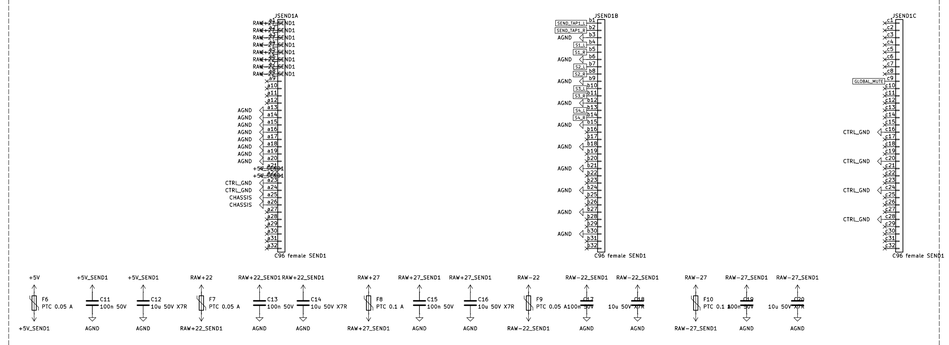


SLOT CH4 (CHANNEL class)

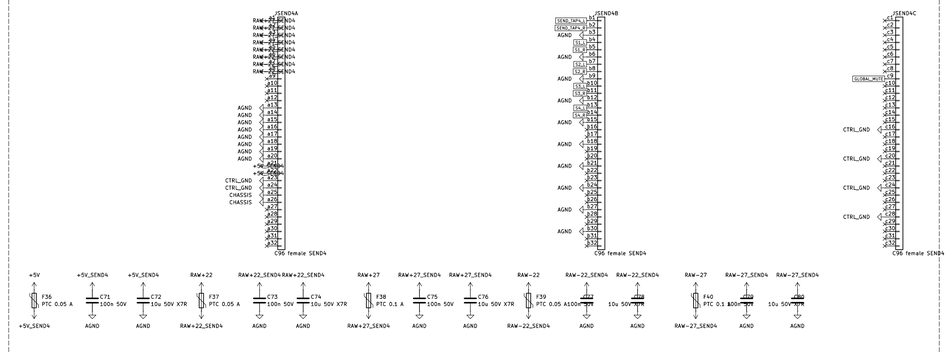
SLOT FIL1 (FILTER class)



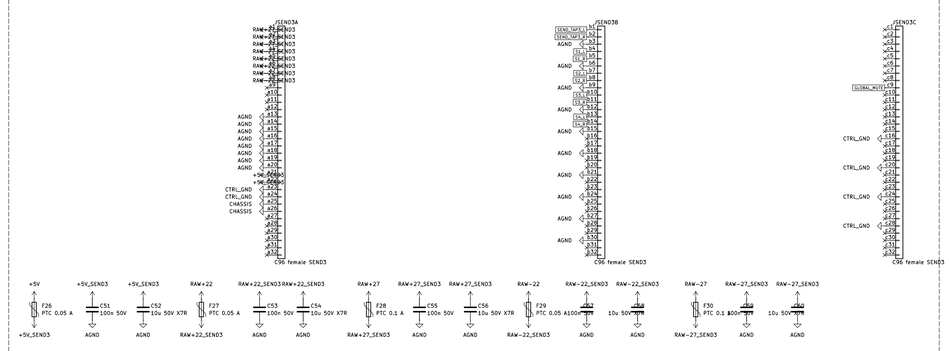
SLOT SEND1 (FX-SEND class)



SLOT SEND2 (FX-SEND class)

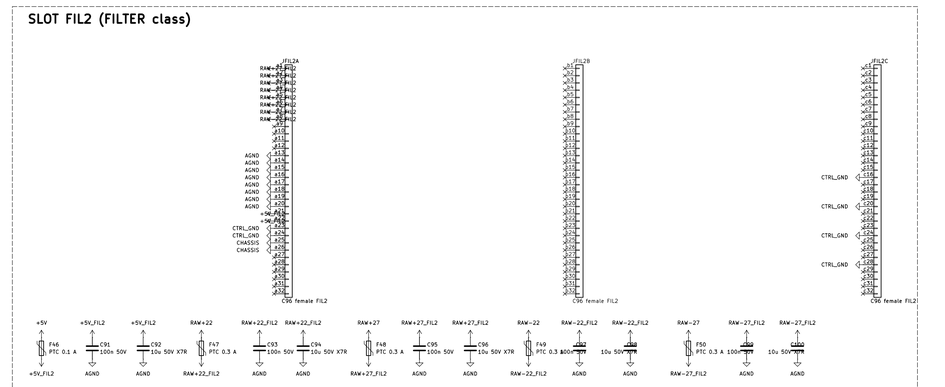


SLOT SEND3 (FX-SEND class)

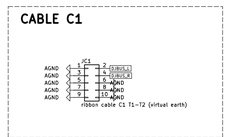


SLOT SEND4 (FX-SEND class)

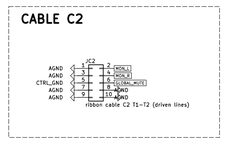
SLOT FIL2 (FILTER class)



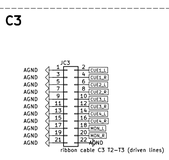
CABLE C1



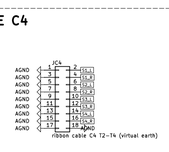
CABLE C2



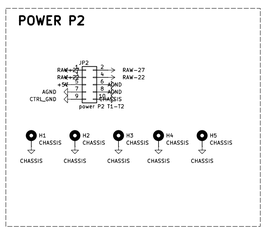
CABLE C3



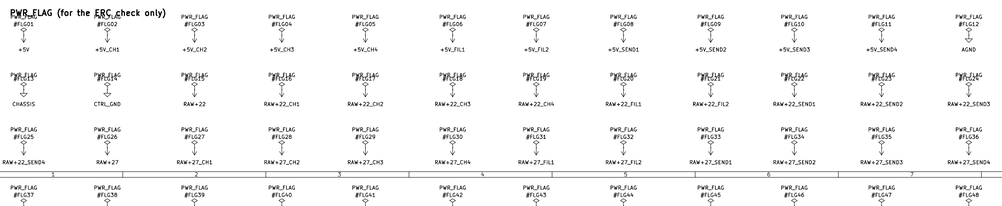
CABLE C4



POWER P2



PWR_FLAG (for the ERC check only)



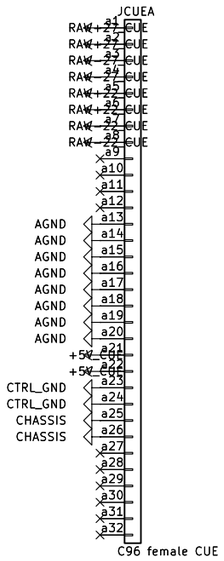
Generated by pcb/tools/legglible.py from the same netlist as the PCB: do not edit by hand

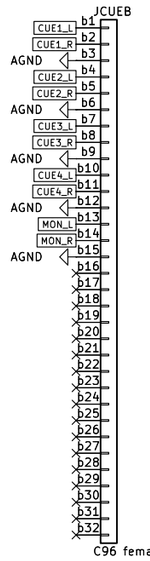
Sheet: /
File: backplane-t2.kicad_sch

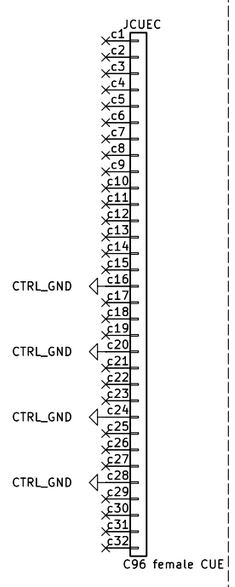
Title: maelcum backplane T2 (channels: FILTER, 4 CHANNELS, 4 FX-SEND) –

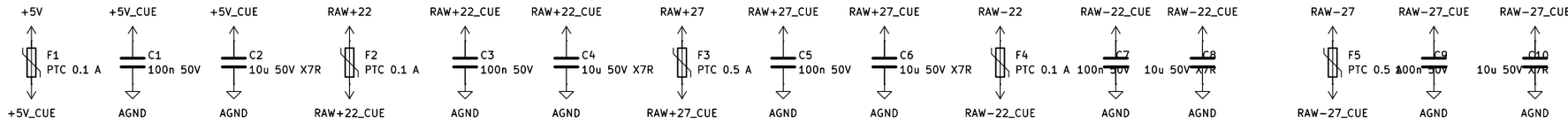
Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

SLOT CUE (CUE class)

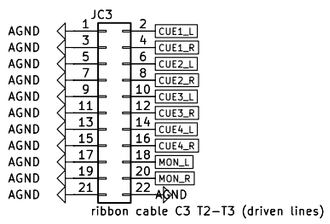




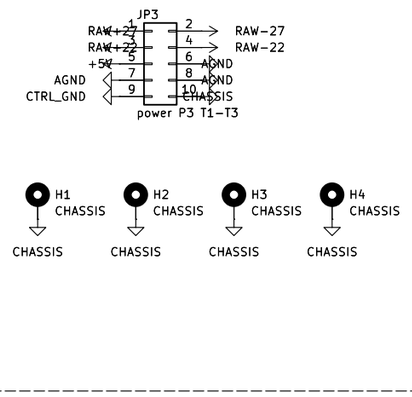




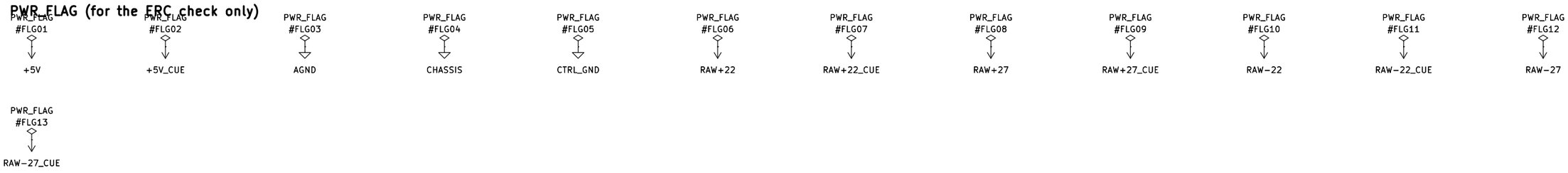
CABLE C3



POWER P3



PWR_FLAG (for the ERC check only)



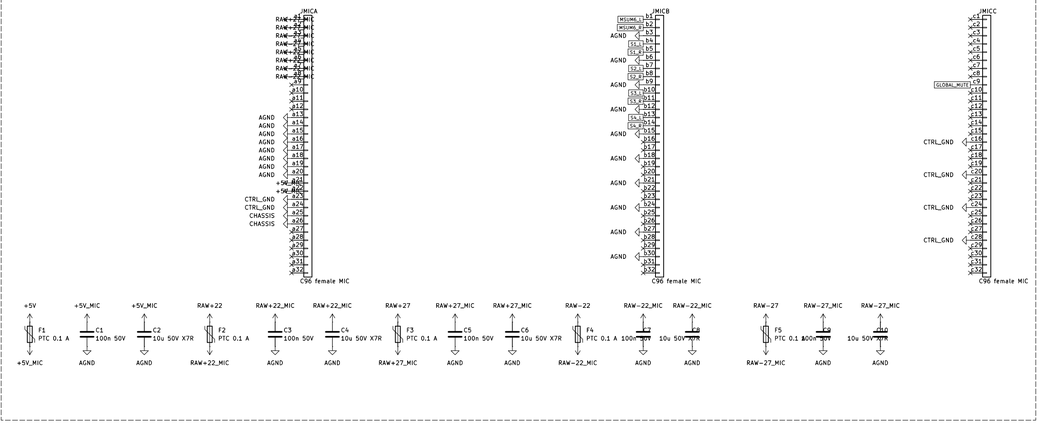
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: backplane-t3.kicad_sch

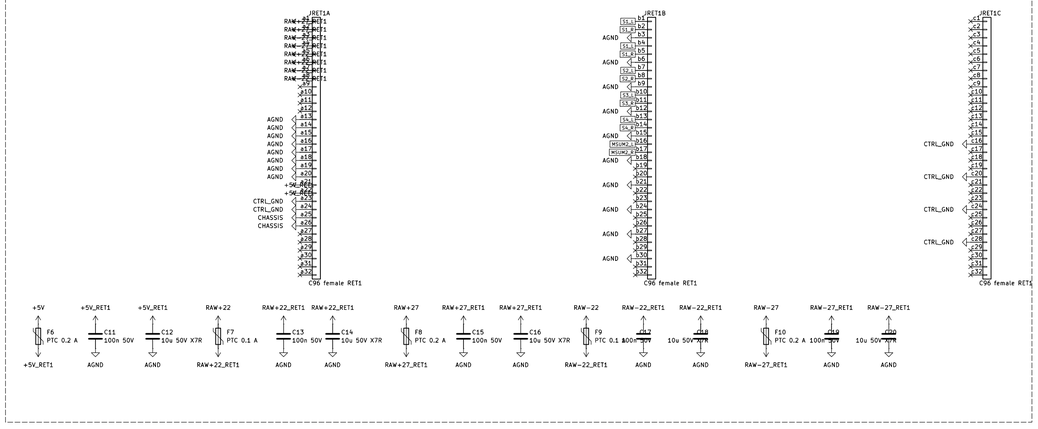
Title: maelcum backplane T3 (headphones: CUE) — draft v0.1 — readable sch

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

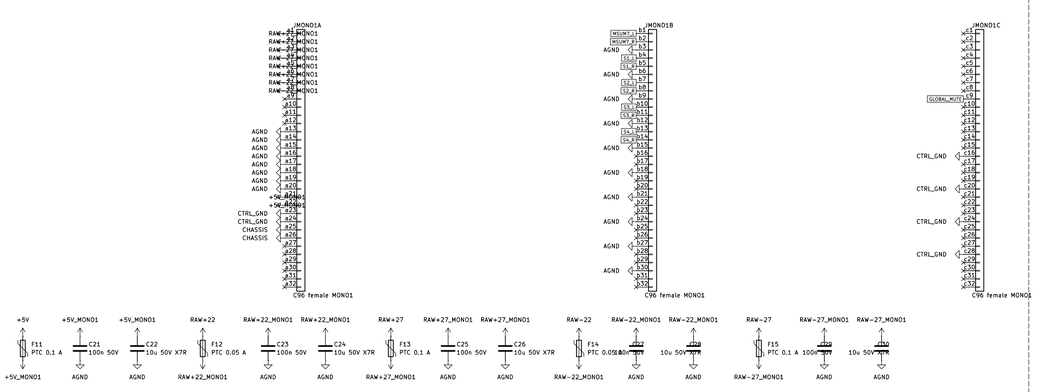
SLOT MIC (MIC class)



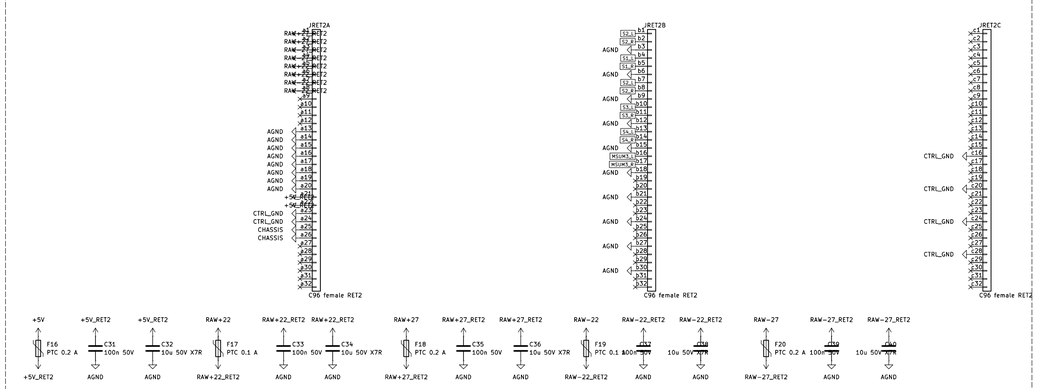
SLOT RET1 (FX–RET class)



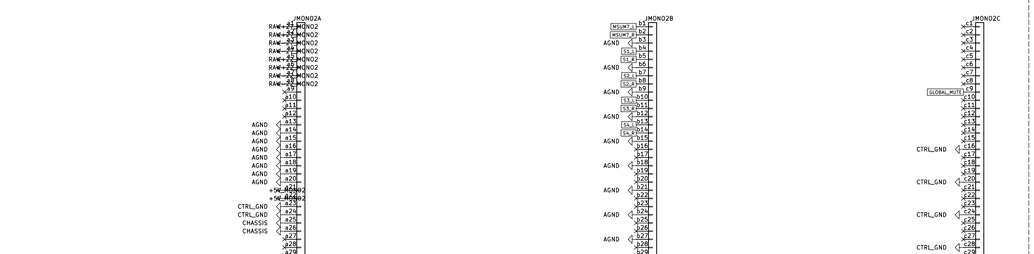
SLOT MONO1 (MONO class)



SLOT RET2 (FX–RET class)



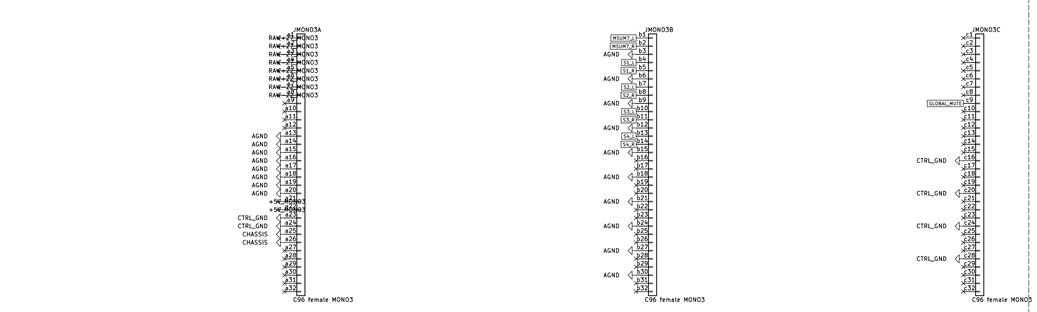
SLOT MONO2 (MONO class)



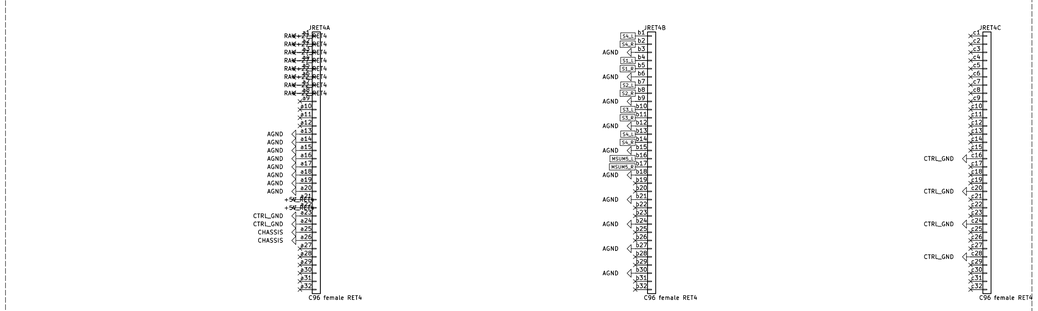
SLOT RET3 (FX–RET class)



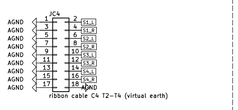
SLOT MONO3 (MONO class)



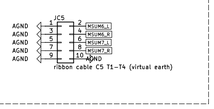
SLOT RET4 (FX–RET class)



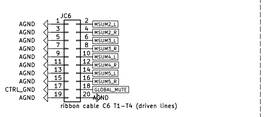
CABLE C4



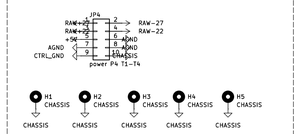
CABLE C5



CABLE C6



POWER P4



PWR_FLAG (for the ERC check only)

Generated by pcb/tools/leggible.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: backplane-t4.kicad_sch

Title: maelcum backplane T4 (right: MIC, 3 MONO, 4 FX–RET) — draft v0.1 —

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1